Real Analysis

Mihir Rao

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Lecture 1: Sets, Set Operations, and Mathematical Induction

The course has two aims: to gain experience with proofs, and to prove statements about real numbers, functions, and limits. This first unit sets up the language of sets and the first proof method, induction.

1.1 Sets and notation

A set is a collection of objects, called its elements or members. The empty set $\emptyset$ has no elements. For a set S, $a \in S$ means that a is an element of S and $a \notin S$ that it is not. Throughout, $\forall$ means "for all", $\exists$ "there exists", $\exists!$ "there exists a unique", := "is defined as", $\implies$ "implies", and $\iff$ "if and only if".

Definition 1.1 [Set relations]. Let A and B be sets.

1. A is a subset of B, written $A \subset B$, if every element of A is in B: $a \in A \implies a \in B$.
2. A and B are equal, $A = B$, if $A \subset B$ and $B \subset A$.
3. A is a proper subset of B, written $A \subsetneq B$, if $A \subset B$ and $A \neq B$.

So $\subset$ allows equality, and proving $A = B$ means proving two inclusions. In set-builder notation, $\{x \in A \mid P(x)\}$, or $\{x \mid P(x)\}$, is the set of all x (in A) satisfying the property $P(x)$; for example $\{x \mid x \text{ is an even number}\}$. The basic sets of the course are

$$\mathbb{N} = \{1, 2, 3, \ldots\}, \qquad \mathbb{Z} = \{0, 1, -1, 2, -2, \ldots\},$$
$$\mathbb{Q} = \left\{ \frac{m}{n} \;\middle|\; m, n \in \mathbb{Z},\ n \neq 0 \right\}, \qquad \mathbb{R},$$

with $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Note that $0 \notin \mathbb{N}$. What $\mathbb{R}$ actually is will be made precise in Lectures 3 and 4.

1.2 Set operations and De Morgan's laws

Definition 1.2 [Set operations]. For sets A and B:

- the union is $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$;
- the intersection is $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$;
- the set difference is $A \setminus B = \{x \in A \mid x \notin B\}$;
- the complement is $A^c = \{x \mid x \notin A\}$;

- A and B are DISJOINT if $A \cap B = \emptyset$.

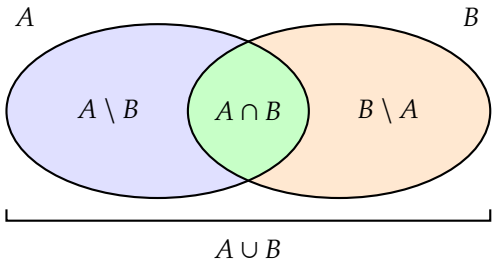


Figure 1: Two overlapping sets. The union $A \cup B$ splits into the three disjoint pieces $A \setminus B$, $A \cap B$, and $B \setminus A$.

Theorem 1.3 [DE MORGAN'S LAWS]. *For sets A, B, C,*

$$(B \cup C)^c = B^c \cap C^c, \qquad A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C),$$
$$(B \cap C)^c = B^c \cup C^c, \qquad A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

Proof. For the first law we prove both inclusions. If $x \in (B \cup C)^c$, then $x \notin B \cup C$, so $x \notin B$ and $x \notin C$; that is, $x \in B^c$ and $x \in C^c$, so $x \in B^c \cap C^c$. Conversely, if $x \in B^c \cap C^c$, then $x \notin B$ and $x \notin C$, so $x \notin B \cup C$ and $x \in (B \cup C)^c$.
The relative version is the same argument inside A: $x \in A \setminus (B \cup C)$ if and only if $x \in A$, $x \notin B$, and $x \notin C$, if and only if $x \in (A \setminus B) \cap (A \setminus C)$. The laws for intersections follow in the same way, with "and" and "or" exchanged. □

1.3 Well-ordering and induction

The natural numbers come with the order $1 < 2 < 3 < \cdots$. Their basic property is taken as an axiom, without proof.

Definition 1.4 [WELL-ORDERING PROPERTY]. The WELL-ORDERING PROPERTY of $\mathbb{N}$: every nonempty $S \subset \mathbb{N}$ has a least element, that is, some $x \in S$ with $x \leq y$ for all $y \in S$.

Theorem 1.5 [INDUCTION]. *Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. Suppose*

1. *(base case) $P(1)$ is true, and*
2. *(inductive step) for each $m \in \mathbb{N}$, if $P(m)$ is true then $P(m+1)$ is true.*

Then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof. Let $S = \{n \in \mathbb{N} \mid P(n) \text{ is false}\}$; we show $S = \emptyset$ by contradiction: assume $S \neq \emptyset$ and derive a false statement. By well-ordering, S has a least element m. Since $P(1)$ is true, $m \neq 1$, so $m > 1$ and $m - 1 \in \mathbb{N}$. As m is least, $m - 1 \notin S$, so $P(m-1)$ is true. The inductive step then makes $P(m)$ true, so $m \notin S$, contradicting $m \in S$. Hence $S = \emptyset$. □

Theorem 1.6 [Geometric sum]. *For every real $c \neq 1$ and every $n \in \mathbb{N}$,*

$$1 + c + c^2 + \cdots + c^n = \frac{1 - c^{n+1}}{1 - c}.$$

Proof. Induct on n. For $n = 1$ the right side is $\frac{1-c^2}{1-c} = \frac{(1-c)(1+c)}{1-c} = 1 + c$. If the formula holds for k, then

$$1 + c + \cdots + c^k + c^{k+1} = \frac{1 - c^{k+1}}{1 - c} + c^{k+1} = \frac{1 - c^{k+1} + c^{k+1}(1 - c)}{1 - c} = \frac{1 - c^{k+2}}{1 - c},$$

which is the formula for $k + 1$. □

Theorem 1.7 [Bernoulli's inequality]. *For every real $c \geq -1$ and every $n \in \mathbb{N}$, $(1 + c)^n \geq 1 + nc$.*

Proof. For $n = 1$ both sides equal $1 + c$. If $(1 + c)^m \geq 1 + mc$, multiply by $1 + c$, which is nonnegative because $c \geq -1$, so the inequality is preserved:

$$(1 + c)^{m+1} = (1 + c)^m(1 + c) \geq (1 + mc)(1 + c) = 1 + (m + 1)c + mc^2 \geq 1 + (m + 1)c.$$

By induction the inequality holds for all n. □

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 1, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 2: Cantor's Theory of Cardinality (Size)

2.1 Functions, images, and preimages

Definition 2.1 [Function]. For sets A and B, a function $f : A \to B$ assigns to each $x \in A$ a unique element $f(x) \in B$. For $C \subset A$ and $D \subset B$, the image of C and the preimage of D are

$$f(C) := \{y \in B \mid y = f(x) \text{ for some } x \in C\}, \qquad f^{-1}(D) := \{x \in A \mid f(x) \in D\}.$$

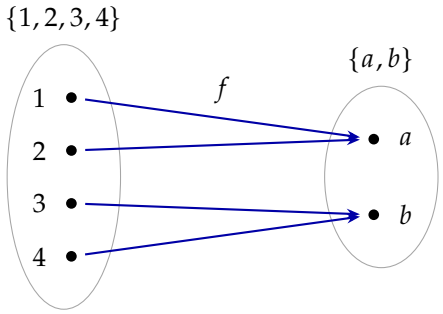


Figure 2: The function $f : \{1, 2, 3, 4\} \to \{a, b\}$ with $f(1) = f(2) = a$ and $f(3) = f(4) = b$. Here $f(\{1, 2\}) = \{a\}$ and $f^{-1}(\{b\}) = \{3, 4\}$.

Definition 2.2 [Injective, surjective, bijective]. A function $f : A \to B$ is

- injective (one-to-one, 1-1) if $f(x_1) = f(x_2)$ implies $x_1 = x_2$;
- surjective (onto) if $f(A) = B$;
- bijective if it is both injective and surjective.

The function in the figure is surjective but not injective. If f is bijective, its inverse $f^{-1} : B \to A$ sends each $y \in B$ to the unique $x \in A$ with $f(x) = y$; thus $f(f^{-1}(y)) = y$ for all $y \in B$.

2.2 Cardinality

When do two sets have the same size? Cantor's answer: when the elements of one can be paired off with the elements of the other.

Definition 2.3 [Cardinality]. Sets A and B have the same cardinality, written $|A| = |B|$, if there is a bijection $f : A \to B$. Further,

- $|A| = n$ if $|A| = |\{1, \ldots, n\}|$, and then A is finite;

- $|A| \le |B|$ if there is an injection $f : A \to B$;
- $|A| < |B|$ if $|A| \le |B|$ but $|A| \neq |B|$.

The notation $\le$ is justified by the following theorem, stated without proof.

Theorem 2.4 [Cantor–Schröder–Bernstein]. *If $|A| \le |B|$ and $|B| \le |A|$, then $|A| = |B|$.*

Definition 2.5 [Countable]. A set A is countably infinite if $|A| = |\mathbb{N}|$, and countable if it is finite or countably infinite. Otherwise A is uncountable.

Proposition 2.6 [Countably infinite sets]. *The even numbers, the odd numbers, and the positive rationals are all countably infinite:*

$$|\{2n \mid n \in \mathbb{N}\}| = |\mathbb{N}|, \qquad |\{2n-1 \mid n \in \mathbb{N}\}| = |\mathbb{N}|, \qquad |\{x \in \mathbb{Q} \mid x > 0\}| = |\mathbb{N}|.$$

Proof. Define $f : \mathbb{N} \to \{2n \mid n \in \mathbb{N}\}$ by $f(n) = 2n$. It is injective, since $2n = 2m$ implies $n = m$, and surjective, since every element of the target has the form $m = 2n = f(n)$. So f is a bijection. For the odd numbers the same argument applies to $g(n) = 2n - 1$. The statement about the positive rationals is recorded here without proof. □

The first two statements show that a proper subset of an infinite set can have the same cardinality as the whole set: in Feynman's phrase, "there are twice as many numbers as numbers."

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 2, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 3: Cantor's Remarkable Theorem and the Rationals' Lack of the Least Upper Bound Property

3.1 Cantor's theorem

Is any set bigger than $\mathbb{N}$? The power set of a set A is the set of all its subsets, $\mathcal{P}(A) = \{B \mid B \subset A\}$. For example,

$$\mathcal{P}(\emptyset) = \{\emptyset\}, \qquad \mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}, \qquad \mathcal{P}(\{1,2\}) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}.$$

In general, if $|A| = n$ then $|\mathcal{P}(A)| = 2^n$, which explains the name.

Theorem 3.1 [Cantor]. *For every set A, $|A| < |\mathcal{P}(A)|$.*

Proof. The map $f : A \to \mathcal{P}(A)$, $f(x) = \{x\}$, is injective, since $\{x\} = \{y\}$ implies $x = y$. So $|A| \leq |\mathcal{P}(A)|$.
Suppose, for contradiction, that $|A| = |\mathcal{P}(A)|$. Then there is a bijection, in particular a surjection, $g : A \to \mathcal{P}(A)$. Let

$$B := \{x \in A \mid x \notin g(x)\} \in \mathcal{P}(A).$$

By surjectivity, $B = g(b)$ for some $b \in A$. Consider two cases.

- If $b \in B$, then by the definition of B, $b \notin g(b) = B$.
- If $b \notin B$, then $b \notin g(b)$, so by definition $b \in B$.

Both cases are contradictory, so no such g exists and $|A| \neq |\mathcal{P}(A)|$. □

Iterating gives an infinite chain of infinite sizes,

$$|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \cdots.$$

Applied to $A = \{1, \ldots, n\}$, and to $A = \emptyset$, Cantor's theorem also gives the following.

Corollary 3.2. *For all $n \in \mathbb{N} \cup \{0\}$, $n < 2^n$.*

3.2 Ordered sets, bounds, and suprema

The goal of the next two lectures is the following characterization: the real numbers have all the algebraic and order properties of the rationals, but none of the holes.

Theorem 3.3 [Real numbers]. *There exists a unique ordered field containing $\mathbb{Q}$ with the least upper bound property. This field is denoted $\mathbb{R}$.*

Ordered fields are defined in Lecture 4; here we define the order structure and the least upper bound property.

Definition 3.4 [Ordered set]. An ordered set is a set S with a relation $<$, called an ordering, such that

1. for all $x, y \in S$, exactly one of $x < y$, $y < x$, $x = y$ holds;
2. if $x < y$ and $y < z$, then $x < z$.

Examples: $\mathbb{Z}$ with $m > n \iff m - n \in \mathbb{N}$; $\mathbb{Q}$ with $p > q$ if and only if $p - q = m/n$ for some $m, n \in \mathbb{N}$; and $\mathbb{Q} \times \mathbb{Q}$ with the dictionary order, $(q, r) > (s, t)$ if $q > s$, or $q = s$ and $r > t$. A non-example is $\mathcal{P}(\mathbb{N})$ ordered by inclusion: $\{1\}$ and $\{2\}$ are not comparable, so the first property fails.

Definition 3.5 [Bounds, supremum, infimum]. Let S be an ordered set and $E \subset S$.

- If some $b \in S$ satisfies $x \leq b$ for all $x \in E$, then E is bounded above and b is an upper bound of E.
- If some $c \in S$ satisfies $x \geq c$ for all $x \in E$, then E is bounded below and c is a lower bound of E.
- $b_0 \in S$ is the least upper bound, or supremum, of E, written $b_0 = \sup E$, if b_0 is an upper bound of E and $b_0 \leq b$ for every upper bound b of E.
- $c_0 \in S$ is the greatest lower bound, or infimum, of E, written $c_0 = \inf E$, if c_0 is a lower bound of E and $c \leq c_0$ for every lower bound c of E.

Example: *Suprema and infima.*

- In $S = \mathbb{Z}$, $E = \{-2, -1, 0, 1, 2\}$ has $\inf E = -2$ and $\sup E = 2$.
- The supremum and infimum need not belong to E: in $S = \mathbb{Q}$, $E = \{q \in \mathbb{Q} \mid 0 \leq q < 1\}$ has $\inf E = 0 \in E$ but $\sup E = 1 \notin E$.
- They need not exist: in $S = \mathbb{Z}$, $E = \mathbb{N}$ has $\inf E = 1$, but no integer is at least every natural number, so $\sup E$ does not exist.

Definition 3.6 [Least upper bound property]. An ordered set S has the least upper bound property if every nonempty $E \subset S$ that is bounded above has a supremum in S.

For example, $-\mathbb{N} = \{-1, -2, -3, \ldots\}$ has the least upper bound property. If $E \subset -\mathbb{N}$ is nonempty,

then $-E \subset \mathbb{N}$ is nonempty, so by well-ordering it has a least element x, and then $-x = \sup E$.

3.3 The rationals lack the least upper bound property

Let $E = \{q \in \mathbb{Q} \mid q > 0,\ q^2 < 2\}$. It is nonempty ($1 \in E$) and bounded above (by 2, since $q \geq 2$ gives $q^2 \geq 4$). We show that a rational supremum would have to be a square root of 2, and that no rational number squares to 2.

Theorem 3.7. *If $x \in \mathbb{Q}$ and $x = \sup E$, then $x > 0$ and $x^2 = 2$.*

Proof. Since $1 \in E$ and x is an upper bound, $x \geq 1 > 0$.
$x^2 \geq 2$. Suppose $x^2 < 2$ and set

$$h = \min\left\{\frac{1}{2},\ \frac{2 - x^2}{2(2x+1)}\right\},$$

so $0 < h < 1$ and $h \in \mathbb{Q}$. Since $h^2 < h$,

$$(x+h)^2 = x^2 + 2xh + h^2 < x^2 + h(2x+1) \leq x^2 + \frac{2-x^2}{2} = \frac{2+x^2}{2} < 2.$$

So $x + h \in E$ and $x + h > x$, contradicting that x is an upper bound.
$x^2 \leq 2$. Suppose $x^2 > 2$ and set $h = \frac{x^2-2}{2x} \in \mathbb{Q}$. Then $h > 0$ and $x - h = \frac{x^2+2}{2x} > 0$, and

$$(x-h)^2 = x^2 - 2xh + h^2 = x^2 - (x^2 - 2) + h^2 = 2 + h^2 > 2.$$

For $q \in E$ this gives $(x-h)^2 - q^2 > 0$, that is, $\big((x-h)+q\big)\big((x-h)-q\big) > 0$. The first factor is positive, so $q < x - h$. Thus $x - h$ is an upper bound of E smaller than x, contradicting $x = \sup E$. Hence $x^2 = 2$. □

Theorem 3.8. *The set $E = \{q \in \mathbb{Q} \mid q > 0,\ q^2 < 2\}$ has no supremum in $\mathbb{Q}$. Hence $\mathbb{Q}$ does not have the least upper bound property.*

Proof. Suppose $x = \sup E \in \mathbb{Q}$. By Theorem 3.7, $x > 0$ and $x^2 = 2$. Then $1 < x < 2$: if $x \leq 1$ then $x^2 \leq 1 < 2$, and if $x \geq 2$ then $x^2 \geq 4 > 2$. Write $x = m/n$ with $m, n \in \mathbb{N}$, so the set

$$S = \{k \in \mathbb{N} \mid kx \in \mathbb{N}\}$$

contains n and is nonempty. By well-ordering it has a least element k_0. Let $k_1 = k_0 x - k_0 = k_0(x-1)$. Since $k_0 x \in \mathbb{N}$, k_1 is an integer; since $1 < x < 2$, we have $0 < k_1 < k_0$, so $k_1 \in \mathbb{N}$ and, by minimality of k_0, $k_1 \notin S$. But

$$k_1 x = k_0 x^2 - k_0 x = 2k_0 - k_0 x = k_0 - k_1 \in \mathbb{N},$$

the last because $k_1 < k_0$. So $k_1 \in S$, a contradiction. $\square$

The second proof uses only $x > 0$ and $x^2 = 2$, so it also shows that no rational number has square 2.

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 3, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 4: The Characterization of the Real Numbers

4.1 Fields

Definition 4.1 [Field]. A set F is a field if it has two operations, addition $+$ and multiplication $\cdot$, with the following properties.

(A1) If $x, y \in F$, then $x + y \in F$.

(A2) (Commutativity) $x + y = y + x$ for all $x, y \in F$.

(A3) (Associativity) $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.

(A4) There is an element $0 \in F$ with $0 + x = x = x + 0$ for all $x \in F$.

(A5) For every $x \in F$ there is $y \in F$ with $x + y = 0$; write $y = -x$.

(M1) If $x, y \in F$, then $x \cdot y \in F$.

(M2) (Commutativity) $x \cdot y = y \cdot x$ for all $x, y \in F$.

(M3) (Associativity) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.

(M4) There is an element $1 \in F$ with $1 \cdot x = x = x \cdot 1$ for all $x \in F$.

(M5) For every $x \in F \setminus \{0\}$ there is $x^{-1} \in F$ with $x \cdot x^{-1} = 1$.

(D) (Distributivity) $(x + y)z = xz + yz$ for all $x, y, z \in F$.

$\mathbb{Q}$ is a field. $\mathbb{Z}$ is not: it fails (M5), since, for example, 2 has no multiplicative inverse in $\mathbb{Z}$.

Example: *Finite fields.* $\mathbb{Z}_2 = \{0, 1\}$ with $1 + 1 = 0$ is a field. So is $\mathbb{Z}_3 = \{0, 1, 2\}$ with addition and multiplication modulo 3; for instance $2 + 1 = 3 \equiv 0$ and $2 \cdot 2 = 4 \equiv 1 \pmod 3$.

The familiar algebraic rules follow from the axioms. For example:

Proposition 4.2. *In any field F, $0 \cdot x = 0$ for all $x \in F$.*

Proof. By (A4) and (D), $0 \cdot x = (0 + 0) \cdot x = 0 \cdot x + 0 \cdot x$. Adding $-(0 \cdot x)$ to both sides gives $0 = 0 \cdot x$. □

4.2 Ordered fields

Definition 4.3 [Ordered field]. A field F is an ordered field if it is also an ordered set with ordering $<$ such that

(i) for all $x, y, z \in F$, $x < y$ implies $x + z < y + z$;

(ii) if $x > 0$ and $y > 0$, then $xy > 0$.

If $x > 0$, then x is POSITIVE; if $x \geq 0$, then x is NONNEGATIVE.

$\mathbb{Q}$ is an ordered field. $\mathbb{Z}_2$ cannot be ordered to make an ordered field. If $0 < 1$, then (i) gives $1 = 1 + 0 < 1 + 1 = 0$, so $1 < 0$, a contradiction. If $1 < 0$, then (i) gives $0 = 1 + 1 < 0 + 1 = 1$, again a contradiction.

All the usual rules for inequalities follow from the ordered field axioms; two examples:

Proposition 4.4. *In an ordered field, $x > 0$ if and only if $-x < 0$.*

Proof. If $x > 0$, adding $-x$ to both sides by (i) gives $0 > -x$. Conversely, if $-x < 0$, adding x gives $0 < x$. □

Proposition 4.5. *In an ordered field, if one of x, y is positive and the other is negative, then $xy < 0$.*

Proof. Suppose $x > 0$ and $y < 0$. Then $-y > 0$ by Proposition 4.4, so $-(xy) = x(-y) > 0$ by (ii), and therefore $xy < 0$ by Proposition 4.4 again. The case $x < 0 < y$ is symmetric: $-(xy) = (-x)y > 0$. Here $-(xy) = x(-y)$ because $xy + x(-y) = x(y + (-y)) = x \cdot 0 = 0$, by distributivity and Proposition 4.2. □

4.3 The greatest lower bound property

In an ordered field, the least upper bound property also gives infima.

Theorem 4.6 [GREATEST LOWER BOUND PROPERTY]. *Let F be an ordered field with the least upper bound property. If $A \subset F$ is nonempty and bounded below, then* $\inf A$ *exists in F.*

Proof. Reflect A: let $B = \{-x \mid x \in A\}$. If a is a lower bound of A, then $a \leq x$ gives $-x \leq -a$ for all $x \in A$, so $-a$ is an upper bound of B. Thus B is nonempty and bounded above, and $c = \sup B$ exists in F.
For all $x \in A$, $-x \leq c$, so $-c \leq x$: $-c$ is a lower bound of A. If a is any lower bound of A, then $-a$ is an upper bound of B, so $c \leq -a$ and hence $a \leq -c$. Therefore $-c = \inf A$. □

4.4 The real numbers

Theorem 4.7 [Real numbers]. *There exists a unique (in a sense that can be made precise) ordered field $\mathbb{R}$ with $\mathbb{Q} \subset \mathbb{R}$ and the least upper bound property.*

$\mathbb{R}$ can be constructed using Dedekind cuts, or as equivalence classes of Cauchy sequences; the theorem is used here without proof. The least upper bound property fills the hole in $\mathbb{Q}$ found in Lecture 3.

Theorem 4.8 [Square root of two]. *There is a unique $r \in \mathbb{R}$ with $r > 0$ and $r^2 = 2$. In other words, $\sqrt{2} \in \mathbb{R}$ but $\sqrt{2} \notin \mathbb{Q}$.*

Proof. Existence. Let $\tilde{E} = \{x \in \mathbb{R} \mid x > 0,\ x^2 < 2\}$. It contains 1 and is bounded above by 2, so $r := \sup \tilde{E}$ exists in $\mathbb{R}$. The proof that a rational supremum of $\{q > 0 : q^2 < 2\}$ has square 2 (Lecture 3) uses only ordered field arithmetic, so it applies verbatim to $\tilde{E}$ in $\mathbb{R}$ and gives $r > 0$ and $r^2 = 2$. By Lecture 3, no rational number has square 2, so $r \notin \mathbb{Q}$.
Uniqueness. If also $\tilde{r} > 0$ and $\tilde{r}^2 = 2$, then $0 = r^2 - \tilde{r}^2 = (r + \tilde{r})(r - \tilde{r})$. Since $r + \tilde{r} > 0$, it has an inverse; multiplying by it gives $r - \tilde{r} = 0$, so $r = \tilde{r}$. □

In the same way one can show that $\sqrt[3]{2} \in \mathbb{R}$.

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Lecture 5: The Archimedean Property, Density of the Rationals, and Absolute Value

5.1 The Archimedean property and density of $\mathbb{Q}$

Between any two reals $x < y$ lies another real, for instance $\frac{x+y}{2}$. The least upper bound property shows that a rational number also lies between them.

Theorem 5.1.

(i) *(Archimedean property) If $x, y \in \mathbb{R}$ and $x > 0$, then there exists $n \in \mathbb{N}$ with $nx > y$.*

(ii) *(Density of $\mathbb{Q}$) If $x, y \in \mathbb{R}$ and $x < y$, then there exists $r \in \mathbb{Q}$ with $x < r < y$.*

Proof. (i) We need $n > y/x$. Suppose instead that $n \leq y/x$ for all $n \in \mathbb{N}$. Then $\mathbb{N}$ is nonempty and bounded above, so $a = \sup \mathbb{N}$ exists in $\mathbb{R}$. Since $a - 1 < a$ is not an upper bound, there is $m \in \mathbb{N}$ with $a - 1 < m$. But then $a < m + 1 \in \mathbb{N}$, so a is not an upper bound of $\mathbb{N}$, a contradiction.
(ii) There are three cases: $0 \leq x < y$, $x < 0 < y$, and $x < y \leq 0$. If $x < 0 < y$, take $r = 0$.
Suppose $0 \leq x < y$. By (i) there is $n \in \mathbb{N}$ with $n(y - x) > 1$, and, again by (i), some $l \in \mathbb{N}$ with $l > nx$. So $S = \{k \in \mathbb{N} \mid k > nx\}$ is nonempty, and by well-ordering it has a least element m. Then $nx < m$. Also $m - 1 \leq nx$: either $m = 1$ and $m - 1 = 0 \leq nx$, or $m - 1 \in \mathbb{N}$ and $m - 1 \notin S$ by minimality. Hence

$$nx < m \leq nx + 1 < ny, \qquad \text{so} \qquad x < \frac{m}{n} < y.$$

If $x < y \leq 0$, then $0 \leq -y < -x$, so the first case gives $\tilde{r} \in \mathbb{Q}$ with $-y < \tilde{r} < -x$, and then $x < -\tilde{r} < y$. □

Proposition 5.2. $\sup\{1 - \frac{1}{n} \mid n \in \mathbb{N}\} = 1$.

Proof. For every $n \in \mathbb{N}$, $1 - \frac{1}{n} < 1$, so 1 is an upper bound. Let x be any upper bound and suppose $x < 1$. By the Archimedean property there is $n \in \mathbb{N}$ with $n(1 - x) > 1$, that is, $x < 1 - \frac{1}{n}$. Then x is not an upper bound, a contradiction. So every upper bound satisfies $x \geq 1$, and the supremum is 1. □

5.2 Working with suprema

The following characterization is the standard way to verify that a number is a supremum.

Theorem 5.3 [ϵ-characterization of the supremum]. *Let $S \subset \mathbb{R}$ be nonempty and bounded above. Then $x = \sup S$ if and only if*

1. *x is an upper bound of S, and*
2. *for every $\epsilon > 0$ there exists $y \in S$ with $x - \epsilon < y \leq x$.*

Proof. If $x = \sup S$ and $\epsilon > 0$, then $x - \epsilon < x$ is not an upper bound, so some $y \in S$ has $y > x - \epsilon$, and $y \leq x$ because x is an upper bound. Conversely, assume (1) and (2), and let b be an upper bound with $b < x$. Taking $\epsilon = x - b$ in (2) gives $y \in S$ with $y > x - \epsilon = b$, contradicting that b is an upper bound. So $x \leq b$ for every upper bound b, and $x = \sup S$. □

For $x \in \mathbb{R}$ and $A \subset \mathbb{R}$ write $x + A := \{x + a \mid a \in A\}$ and $xA := \{xa \mid a \in A\}$.

Theorem 5.4 [Translation and positive scaling]. *Let $A \subset \mathbb{R}$ be nonempty and bounded above.*

1. *If $x \in \mathbb{R}$, then $x + A$ is bounded above and $\sup(x + A) = x + \sup A$.*
2. *If $x > 0$, then xA is bounded above and $\sup(xA) = x \sup A$.*

Proof. By the least upper bound property, $\sup A \in \mathbb{R}$, and $a \leq \sup A$ for all $a \in A$.
(1) Then $x + a \leq x + \sup A$ for all $a \in A$, so $x + \sup A$ is an upper bound of $x + A$. Given $\epsilon > 0$, Theorem 5.3 gives $y \in A$ with $\sup A - \epsilon < y \leq \sup A$, so $(x + \sup A) - \epsilon < x + y \leq x + \sup A$ with $x + y \in x + A$. By Theorem 5.3, $\sup(x + A) = x + \sup A$.
(2) Since $x > 0$, $xa \leq x \sup A$ for all $a \in A$, so $x \sup A$ is an upper bound of xA. Given $\epsilon > 0$, choose $y \in A$ with $\sup A - \frac{\epsilon}{x} < y \leq \sup A$; multiplying by $x > 0$ gives $x \sup A - \epsilon < xy \leq x \sup A$. By Theorem 5.3, $\sup(xA) = x \sup A$. □

Theorem 5.5. *Let $A, B \subset \mathbb{R}$ be nonempty with $x \leq y$ for all $x \in A$ and $y \in B$. Then $\sup A \leq \inf B$.*

Proof. Every $y \in B$ is an upper bound of A, so $\sup A$ exists and $\sup A \leq y$. Thus $\sup A$ is a lower bound of B, so $\inf B$ exists and $\sup A \leq \inf B$. □

5.3 Absolute value

Definition 5.6 [Absolute value]. For $x \in \mathbb{R}$, the absolute value is

$$|x| := \begin{cases} x, & x \geq 0, \\ -x, & x \leq 0. \end{cases}$$

Theorem 5.7 [Properties of the absolute value]. *For all $x, y \in \mathbb{R}$:*

1. $|x| \geq 0$, *and* $|x| = 0$ *if and only if* $x = 0$;
2. $|-x| = |x|$;
3. $|xy| = |x|\,|y|$;
4. $|x^2| = x^2 = |x|^2$;
5. $|x| \leq y$ *if and only if* $-y \leq x \leq y$;
6. $x \leq |x|$.

Proof. Each item is a check by cases on signs.
(1) If $x \geq 0$, then $|x| = x \geq 0$; if $x \leq 0$, then $|x| = -x \geq 0$. If $x = 0$, then $|x| = 0$. Conversely, if $|x| = 0$, then $x = |x| = 0$ when $x \geq 0$ and $-x = |x| = 0$ when $x \leq 0$; either way $x = 0$.
(2) If $x \geq 0$, then $-x \leq 0$ and $|-x| = -(-x) = x = |x|$. If $x \leq 0$, then $-x \geq 0$ and $|-x| = -x = |x|$.
(3) If $x, y \geq 0$, then $xy \geq 0$ and $|xy| = xy = |x|\,|y|$. If $x, y \leq 0$, then $xy = (-x)(-y) \geq 0$, so $|xy| = (-x)(-y) = |x|\,|y|$. If $x \geq 0 \geq y$, then $xy \leq 0$, so $|xy| = -xy = x(-y) = |x|\,|y|$; the case $x \leq 0 \leq y$ is symmetric.
(4) Take $y = x$ in (3): $|x^2| = |x|^2$. Since $x^2 \geq 0$, $|x^2| = x^2$.
(5) Suppose $|x| \leq y$. If $x \geq 0$, then $-y \leq 0 \leq x = |x| \leq y$. If $x \leq 0$, then $|-x| = |x| \leq y$ with $-x \geq 0$, so the previous case gives $-y \leq -x \leq y$, that is, $-y \leq x \leq y$. Conversely, suppose $-y \leq x \leq y$. If $x \geq 0$, then $|x| = x \leq y$; if $x \leq 0$, then $|x| = -x \leq y$.
(6) Take $y = |x|$ in (5). □

These properties lead to the triangle inequality in the next lecture.

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Lecture 6: The Uncountability of the Real Numbers

6.1 The triangle inequality

Theorem 6.1 [Triangle inequality]. *For all $x, y \in \mathbb{R}$,*

$$|x + y| \le |x| + |y|.$$

Proof. Since $x \le |x|$ and $y \le |y|$, we have $x + y \le |x| + |y|$. Applying the same bound to $-x$ and $-y$ gives $-(x + y) \le |-x| + |-y| = |x| + |y|$. Hence $-(|x| + |y|) \le x + y \le |x| + |y|$, which is equivalent to $|x + y| \le |x| + |y|$ by the characterization $|a| \le b \iff -b \le a \le b$ from Lecture 5. □

6.2 Decimal expansions

A set A is countable if it is finite or $|A| = |\mathbb{N}|$, and $\mathbb{Q}$ is countable. To ask whether $\mathbb{R}$ is countable, represent real numbers by decimal expansions. A rational number has a finite expansion

$$x = 10^k d_k + \cdots + 10 d_1 + d_0 + 10^{-1} d_{-1} + \cdots + 10^{-M} d_{-M}, \qquad d_i \in \{0, 1, \ldots, 9\},$$

written $x = d_k d_{k-1} \cdots d_1 d_0 . d_{-1} \cdots d_{-M}$. Real numbers are obtained by allowing infinitely many digits, with the infinite sum defined as a supremum.

Definition 6.2 [Decimal representation]. Let $x \in (0, 1]$ and let $d_{-j} \in \{0, 1, \ldots, 9\}$ for $j \in \mathbb{N}$. We say that x is represented by the digits $\{d_{-j}\}$, and write $x = 0.d_{-1}d_{-2}\cdots$, if

$$x = \sup\left\{10^{-1} d_{-1} + 10^{-2} d_{-2} + \cdots + 10^{-n} d_{-n} : n \in \mathbb{N}\right\}.$$

For example, the partial sums for $0.2500\cdots$ are $\frac{1}{5}, \frac{1}{4}, \frac{1}{4}, \ldots$, so $0.2500\cdots = \sup\{\frac{1}{5}, \frac{1}{4}\} = \frac{1}{4}$.

Theorem 6.3 [Existence and uniqueness of expansions]. *For every $x \in (0, 1]$ there is a unique sequence of digits $\{d_{-j}\}$ such that $x = 0.d_{-1}d_{-2}\cdots$ and, for every $n \in \mathbb{N}$,*

$$0.d_{-1}d_{-2}\cdots d_{-n} < x \le 0.d_{-1}d_{-2}\cdots d_{-n} + 10^{-n}. \tag{6.1}$$

Conversely, for every sequence of digits $\{d_{-j}\}$ there is a unique $x \in [0, 1]$ with $x = 0.d_{-1}d_{-2}\cdots$.

The strict inequality on the left selects the expansion that does not terminate. For instance, the expansion of $\frac{1}{2}$ in this theorem is $0.4999\cdots$, not $0.5000\cdots$, since $0.5 < \frac{1}{2}$ fails.

6.3 Cantor's theorem

Theorem 6.4 [Cantor]. *The interval* $(0, 1]$ *is uncountable.*

Proof. Suppose, for contradiction, that $(0, 1]$ is countable. It is infinite, so there is a bijection $x : \mathbb{N} \to (0, 1]$. Write the expansion of Theorem 6.3 of each $x(n)$ as

$$x(n) = 0.d_{-1}^{(n)} d_{-2}^{(n)} d_{-3}^{(n)} \cdots ,$$

where the superscript (n) is an index, not an exponent. Change the diagonal digits: set

$$e_{-j} = \begin{cases} 1, & d_{-j}^{(j)} \neq 1, \\ 2, & d_{-j}^{(j)} = 1, \end{cases}$$

and let $y = 0.e_{-1}e_{-2}\cdots$. Then $y \geq 0.1 > 0$ and $y \leq 1$, so $y \in (0, 1]$.
The digits e_{-j} satisfy (6.1) for y. Every digit after position n is positive, so the partial sums strictly exceed $0.e_{-1}\cdots e_{-n}$ from position $n+1$ on, and y is strictly larger. Every digit is at most 2, so every partial sum is at most $0.e_{-1}\cdots e_{-n} + \sum_{j>n} 2 \cdot 10^{-j} = 0.e_{-1}\cdots e_{-n} + \frac{2}{9}\,10^{-n}$, and so is y. By uniqueness in Theorem 6.3, $0.e_{-1}e_{-2}\cdots$ is *the* expansion of y.
For each n, the nth digits of y and $x(n)$ differ: $e_{-n} \neq d_{-n}^{(n)}$. Since the expansions are unique, $y \neq x(n)$ for every n. So y is not in the range of x, contradicting surjectivity. □

$x(1) = 0.$	$d_{-1}^{(1)}$	$d_{-2}^{(1)}$	$d_{-3}^{(1)}$	$d_{-4}^{(1)}$	$\cdots$	
$x(2) = 0.$	$d_{-1}^{(2)}$	$d_{-2}^{(2)}$	$d_{-3}^{(2)}$	$d_{-4}^{(2)}$	$\cdots$	
$x(3) = 0.$	$d_{-1}^{(3)}$	$d_{-2}^{(3)}$	$d_{-3}^{(3)}$	$d_{-4}^{(3)}$	$\cdots$	
$x(4) = 0.$	$d_{-1}^{(4)}$	$d_{-2}^{(4)}$	$d_{-3}^{(4)}$	$d_{-4}^{(4)}$	$\cdots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	
$y = 0.$	e_{-1}	e_{-2}	e_{-3}	e_{-4}	$\cdots$	$e_{-j} \neq d_{-j}^{(j)}$

Figure 3: Cantor's diagonal argument. Row n lists the digits of $x(n)$. The digit e_{-n} of y is chosen to differ from the circled diagonal digit $d_{-n}^{(n)}$, so y differs from every row.

Corollary 6.5. $\mathbb{R}$ *is uncountable.*

Proof. A subset of a countable set is countable, and $(0, 1] \subset \mathbb{R}$ is uncountable. □

6.4 Sequences and convergence

Definition 6.6 [Sequence]. A sequence of real numbers is a function $x : \mathbb{N} \to \mathbb{R}$. We write $x_n = x(n)$ and denote the sequence by $\{x_n\}_{n=1}^{\infty}$, $\{x_n\}$, or $x_1, x_2, \ldots$. A sequence $\{x_n\}$ is bounded if there is $B \geq 0$ with $|x_n| \leq B$ for all n.

For example, $x_n = 1/n$ is bounded by 1, while $x_n = n$ is not bounded. A sequence is not the same as its set of values: the sequence $\{(-1)^n\}_{n=1}^{\infty}$ is $-1, 1, -1, 1, \ldots$, while $\{(-1)^n : n \in \mathbb{N}\} = \{-1, 1\}$.

Definition 6.7 [Convergence]. A sequence $\{x_n\}$ converges to $x \in \mathbb{R}$ if for every $\epsilon > 0$ there is $M \in \mathbb{N}$ such that

$$|x_n - x| < \epsilon \qquad \text{for all } n \geq M.$$

A sequence that converges to some x is convergent; otherwise it is divergent.

Negating the definition, $\{x_n\}$ does not converge to x if there is $\epsilon_0 > 0$ such that for every $M \in \mathbb{N}$ some $n \geq M$ has $|x_n - x| \geq \epsilon_0$. A sequence is divergent if this holds for every $x \in \mathbb{R}$.

Lemma 6.8. *Let $x, y \in \mathbb{R}$. If $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.*

Proof. If $x \neq y$, take $\epsilon = |x - y|/2 > 0$. Then $|x - y| < |x - y|/2$, so $|x - y|/2 < 0$, a contradiction. □

Theorem 6.9 [Uniqueness of limits]. *If $\{x_n\}$ converges to x and to y, then $x = y$.*

Proof. Let $\epsilon > 0$. Choose N_1 with $|x_n - x| < \epsilon/2$ for $n \geq N_1$, and N_2 with $|x_n - y| < \epsilon/2$ for $n \geq N_2$. For $n = \max\{N_1, N_2\}$, the triangle inequality gives

$$|x - y| \leq |x - x_n| + |x_n - y| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon.$$

Since ϵ was arbitrary, $x = y$ by Lemma 6.8. □

So the limit of a convergent sequence is well defined; we write $x = \lim_{n\to\infty} x_n$ or $x_n \to x$.

Example: *Two basic limits.* A constant sequence $x_n = c$ converges to c: take $M = 1$, since $|x_n - c| = 0 < \epsilon$ for all n. Also $\lim_{n\to\infty} 1/n = 0$: given $\epsilon > 0$, choose $M \in \mathbb{N}$ with $M > 1/\epsilon$ by the Archimedean property. Then for $n \geq M$,

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{M} < \epsilon.$$

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Lecture 7: Convergent Sequences of Real Numbers

7.1 Two examples

Example: *A rational function of n.* To show $\lim_{n\to\infty} \frac{1}{n^2+2n+100} = 0$, it suffices to bound the term by something simpler that tends to 0. Given $\epsilon > 0$, choose $M \in \mathbb{N}$ with $M > \frac{1}{2\epsilon}$. For $n \geq M$,

$$\left|\frac{1}{n^2+2n+100} - 0\right| = \frac{1}{n^2+2n+100} \leq \frac{1}{2n} \leq \frac{1}{2M} < \epsilon.$$

Example: *The sequence $(-1)^n$ diverges.* Let $x \in \mathbb{R}$; we show $(-1)^n \not\to x$ with $\epsilon_0 = \frac{1}{2}$. For any $M \in \mathbb{N}$,

$$2 = |(-1)^M - (-1)^{M+1}| \leq |(-1)^M - x| + |(-1)^{M+1} - x|,$$

so $|(-1)^M - x| \geq \frac{1}{2}$ or $|(-1)^{M+1} - x| \geq \frac{1}{2}$. Thus some $n \geq M$ has $|(-1)^n - x| \geq \epsilon_0$. Since x was arbitrary, $\{(-1)^n\}$ is divergent.

7.2 Convergent sequences are bounded

Theorem 7.1. *If $\{x_n\}$ is convergent, then $\{x_n\}$ is bounded.*

Proof. Let $x_n \to x$ and choose $M \in \mathbb{N}$ with $|x_n - x| < 1$ for all $n \geq M$. Set

$$B = \max\{|x_1|, |x_2|, \ldots, |x_{M-1}|, |x| + 1\}.$$

If $n < M$, then $|x_n| \leq B$ by construction. If $n \geq M$, then $|x_n| \leq |x_n - x| + |x| < 1 + |x| \leq B$. □

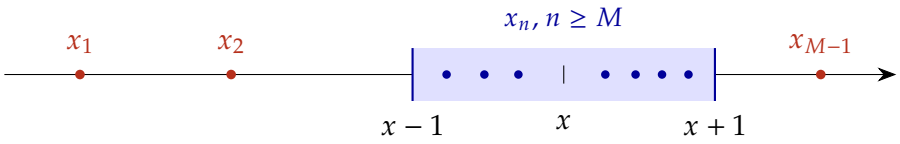


Figure 4: Schematic of the boundedness argument. From index M on, every term lies in $(x-1, x+1)$, so $|x_n| < |x| + 1$; only $x_1, \ldots, x_{M-1}$ can lie outside.

7.3 Monotone sequences

Definition 7.2 [Monotone sequence]**.** A sequence $\{x_n\}$ is monotone increasing if $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$, and monotone decreasing if $x_n \geq x_{n+1}$ for all n. It is monotone if it is either.

For example, $x_n = 1/n$ is monotone decreasing, $y_n = -1/n$ is monotone increasing, and $(-1)^n$ is neither.

Theorem 7.3 [Monotone convergence]. *A monotone increasing sequence $\{x_n\}$ is convergent if and only if it is bounded, and then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Likewise, a monotone decreasing sequence is convergent if and only if it is bounded, and then $\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}$.

Proof. A convergent sequence is bounded by Theorem 7.1. Conversely, let $\{x_n\}$ be increasing and bounded. By the least upper bound property of $\mathbb{R}$, $x := \sup\{x_n : n \in \mathbb{N}\}$ exists. Let $\epsilon > 0$. Since $x - \epsilon$ is not an upper bound, some $M \in \mathbb{N}$ has $x - \epsilon < x_M \le x$. For $n \ge M$, monotonicity gives

$$x - \epsilon < x_M \le x_n \le x < x + \epsilon,$$

so $|x_n - x| < \epsilon$. Hence $x_n \to x$.
For a bounded decreasing sequence, let $x = \inf\{x_n : n \in \mathbb{N}\}$. Given $\epsilon > 0$, some M has $x \le x_M < x + \epsilon$, and for $n \ge M$, $x - \epsilon < x \le x_n \le x_M < x + \epsilon$. □

7.4 Subsequences

Definition 7.4 [Subsequence]. Let $\{x_n\}$ be a sequence and let $\{n_k\}_{k=1}^{\infty}$ be a strictly increasing sequence of natural numbers, $n_1 < n_2 < \cdots$. The sequence $\{x_{n_k}\}_{k=1}^{\infty}$ is a subsequence of $\{x_n\}$.

For $x_n = n$, the sequences $1, 3, 5, 7, \ldots$ (that is, x_{2k-1}), $2, 4, 6, 8, \ldots$ (that is, x_{2k}), and the primes $2, 3, 5, 7, 11, \ldots$ (that is, x_{p_k} with p_k the kth prime) are subsequences. The sequences $1, 1, 1, \ldots$ and $1, 1, 3, 3, 5, 5, \ldots$ are not, since their indices do not strictly increase. For $x_n = (-1)^n$, the subsequence x_{2k-1} is $-1, -1, -1, \ldots$ and the subsequence x_{2k} is $1, 1, 1, \ldots$.

Theorem 7.5. *If $x_n \to x$, then every subsequence $\{x_{n_k}\}$ converges to x.*

Proof. First, $n_k \ge k$ for all k, by induction: $n_1 \ge 1$, and $n_{k+1} > n_k \ge k$ gives $n_{k+1} \ge k + 1$. Let $\epsilon > 0$ and choose M_0 with $|x_n - x| < \epsilon$ for $n \ge M_0$. If $k \ge M_0$, then $n_k \ge k \ge M_0$, so $|x_{n_k} - x| < \epsilon$. □

This gives a second proof that $\{(-1)^n\}$ diverges: if it converged to some x, its subsequences $-1, -1, \ldots$ and $1, 1, \ldots$ would both converge to x, so $x = -1 = 1$.

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Lecture 8: The Squeeze Theorem and Operations Involving Convergent Sequences

8.1 The squeeze theorem

Theorem 8.1 [Squeeze theorem]. *Let $\{a_n\}$, $\{b_n\}$, $\{x_n\}$ be sequences with $a_n \le x_n \le b_n$ for all $n \in \mathbb{N}$. If $\{a_n\}$ and $\{b_n\}$ converge and $\lim a_n = x = \lim b_n$, then $\{x_n\}$ converges and $\lim_{n\to\infty} x_n = x$.*

Proof. Let $\epsilon > 0$. Choose M_0 with $|a_n - x| < \epsilon$, so $x - \epsilon < a_n$, for $n \ge M_0$, and M_1 with $b_n < x + \epsilon$ for $n \ge M_1$. For $n \ge M = \max\{M_0, M_1\}$,

$$x - \epsilon < a_n \le x_n \le b_n < x + \epsilon,$$

so $|x_n - x| < \epsilon$. □

The definition of convergence also gives directly

$$\lim_{n\to\infty} x_n = x \iff \lim_{n\to\infty} |x_n - x| = 0, \tag{8.1}$$

since $\big||x_n - x| - 0\big| = |x_n - x|$. Combined with the squeeze theorem for $0 \le |x_n - x| \le c_n$ with $c_n \to 0$, this is the usual way to prove a limit.

Example: $\lim_{n\to\infty} \frac{n^2}{n^2+n+1} = 1$. We have

$$\left|\frac{n^2}{n^2+n+1} - 1\right| = \frac{n+1}{n^2+n+1} \le \frac{n+1}{n^2+n} = \frac{1}{n}.$$

Since $0 \le |\cdot| \le 1/n \to 0$, the squeeze theorem and (8.1) give the limit.

8.2 Limits and order

Theorem 8.2. *Let $\{x_n\}$ and $\{y_n\}$ be sequences of real numbers.*

1. *If both converge and $x_n \le y_n$ for all n, then $\lim x_n \le \lim y_n$.*
2. *If $\{x_n\}$ converges and $a \le x_n \le b$ for all n, then $a \le \lim x_n \le b$.*

Proof. (1) Let $x = \lim x_n$, $y = \lim y_n$, and suppose $y < x$. Take $\epsilon = \frac{x-y}{2}$ and choose M large

enough that both $|y_M - y| < \epsilon$ and $|x_M - x| < \epsilon$. Then

$$y_M < y + \frac{x-y}{2} = \frac{x+y}{2} = x - \frac{x-y}{2} < x_M,$$

contradicting $x_M \le y_M$.
(2) Apply (1) to the constant sequence a and $\{x_n\}$, and to $\{x_n\}$ and the constant sequence b. □

8.3 Limits and algebraic operations

Theorem 8.3 [Algebra of limits]. *Suppose $x_n \to x$ and $y_n \to y$. Then*

1. $x_n + y_n \to x + y$;
2. $cx_n \to cx$ *for every* $c \in \mathbb{R}$;
3. $x_n y_n \to xy$;
4. *if $y_n \neq 0$ for all n and $y \neq 0$, then $x_n/y_n \to x/y$.*

Proof. (1) Given $\epsilon > 0$, choose M so that $|x_n - x| < \epsilon/2$ and $|y_n - y| < \epsilon/2$ for $n \ge M$. Then $|x_n + y_n - (x+y)| \le |x_n - x| + |y_n - y| < \epsilon$.
(2) Choose M with $|x_n - x| < \epsilon/(|c|+1)$ for $n \ge M$. Then $|cx_n - cx| = |c||x_n - x| < \frac{|c|}{|c|+1}\epsilon < \epsilon$.
(3) By Lecture 7, $\{y_n\}$ is bounded, say $|y_n| \le B$. Then

$$|x_n y_n - xy| = |(x_n - x)y_n + (y_n - y)x| \le B|x_n - x| + |x||y_n - y|.$$

The right side tends to 0 by (1), (2), and (8.1), so $|x_n y_n - xy| \to 0$ by the squeeze theorem.
(4) By (3) it suffices to show $1/y_n \to 1/y$. First, $|y_n|$ is bounded away from 0. Choose M_0 with $|y_n - y| < |y|/2$ for $n \ge M_0$; then $|y| \le |y_n - y| + |y_n| < |y|/2 + |y_n|$, so $|y_n| > |y|/2$. Let

$$b = \min\{|y_1|, \dots, |y_{M_0-1}|, |y|/2\} > 0,$$

so that $|y_n| \ge b$ for all n. Then

$$0 \le \left|\frac{1}{y_n} - \frac{1}{y}\right| = \frac{|y_n - y|}{|y_n||y|} \le \frac{1}{b|y|}|y_n - y| \to 0,$$

and the squeeze theorem gives $1/y_n \to 1/y$. □

By induction with (3), $x_n^k \to x^k$ for every $k \in \mathbb{N}$.

8.4 Square roots and absolute values

Theorem 8.4. *If $x_n \to x$ and $x_n \geq 0$ for all n, then $\sqrt{x_n} \to \sqrt{x}$.*

Proof. By Theorem 8.2, $x \geq 0$.
Case $x = 0$. Given $\epsilon > 0$, choose M with $x_n = |x_n - 0| < \epsilon^2$ for $n \geq M$. Then $|\sqrt{x_n} - 0| = \sqrt{x_n} < \epsilon$.
Case $x > 0$. Multiplying and dividing by $\sqrt{x_n} + \sqrt{x} > 0$,

$$\left|\sqrt{x_n} - \sqrt{x}\right| = \frac{|x_n - x|}{\sqrt{x_n} + \sqrt{x}} \leq \frac{1}{\sqrt{x}}|x_n - x|,$$

and the squeeze theorem gives $|\sqrt{x_n} - \sqrt{x}| \to 0$. □

The cases are needed because the bound in the second case divides by $\sqrt{x}$.

Theorem 8.5. *If $x_n \to x$, then $|x_n| \to |x|$.*

Proof. Since $|t| = \sqrt{t^2}$ for every real t, and $x_n^2 \to x^2$ by Theorem 8.3, Theorem 8.4 gives $|x_n| = \sqrt{x_n^2} \to \sqrt{x^2} = |x|$. □

8.5 Geometric sequences

Theorem 8.6. *If $0 < c < 1$, then $c^n \to 0$. If $c > 1$, then $\{c^n\}$ is unbounded.*

Proof. Let $0 < c < 1$. By induction, $0 < c^{n+1} < c^n < 1$ for all n: multiplying $0 < c^{m+1} < c^m$ by $c > 0$ gives the next step. So $\{c^n\}$ is monotone decreasing and bounded below by 0, and converges to some L by the monotone convergence theorem of Lecture 7. To show $L = 0$, let $\epsilon > 0$ and choose M with $|c^n - L| < (1-c)\epsilon/2$ for $n \geq M$. Then

$$(1-c)|L| = |L - cL| \leq |L - c^{M+1}| + c|c^M - L| < (1-c)\tfrac{\epsilon}{2} + c(1-c)\tfrac{\epsilon}{2} < (1-c)\epsilon.$$

Dividing by $1 - c > 0$, $|L| < \epsilon$ for every $\epsilon > 0$, so $L = 0$.
Let $c > 1$ and $B \geq 0$. Choose $n \in \mathbb{N}$ with $n > B/(c-1)$. By Bernoulli's inequality $(1+t)^n \geq 1 + nt$ for $t \geq -1$ (Lecture 1), with $t = c - 1$,

$$c^n = (1 + (c-1))^n \geq 1 + n(c-1) > n(c-1) > B.$$

So no B bounds $\{c^n\}$. □

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Lecture 9: Limsup, Liminf, and the Bolzano-Weierstrass Theorem

9.1 Some special sequences

Theorem 9.1 [Special sequences]**.**

1. *If* $p > 0$, *then* $\lim_{n\to\infty} n^{-p} = 0$.
2. *If* $p > 0$, *then* $\lim_{n\to\infty} p^{1/n} = 1$.
3. $\lim_{n\to\infty} n^{1/n} = 1$.

Proof. (1) Given $\epsilon > 0$, choose $M > (1/\epsilon)^{1/p}$. For $n \geq M$, $|n^{-p} - 0| = 1/n^p \leq 1/M^p < \epsilon$.
(2) Suppose $p > 1$. Then $p^{1/n} > 1$, since $p^{1/n} \leq 1$ would give $p = (p^{1/n})^n \leq 1$. Bernoulli's inequality gives

$$p = \big(1 + (p^{1/n} - 1)\big)^n \geq 1 + n(p^{1/n} - 1), \qquad \text{so} \qquad 0 < p^{1/n} - 1 \leq \frac{p-1}{n},$$

and $p^{1/n} \to 1$ by the squeeze theorem. If $0 < p < 1$, then $1/p > 1$ and $p^{1/n} = 1/(1/p)^{1/n} \to 1/1 = 1$ by the quotient rule of Lecture 8. If $p = 1$, the sequence is constant.
(3) Let $x_n = n^{1/n} - 1 \geq 0$. For $n > 1$, keeping only the $j = 2$ term of the binomial expansion,

$$n = (1 + x_n)^n = \sum_{j=0}^{n} \binom{n}{j} x_n^j \geq \binom{n}{2} x_n^2 = \frac{n(n-1)}{2} x_n^2,$$

so $0 \leq x_n \leq \sqrt{2/(n-1)}$. The bound tends to 0 by (1) and the square-root rule of Lecture 8, so $x_n \to 0$ and $n^{1/n} \to 1$. □

9.2 Limit superior and limit inferior

The goal is to show that every bounded sequence has a convergent subsequence. The candidates for its limit are the limit superior and limit inferior.

Definition 9.2 [Limit superior and inferior]**.** Let $\{x_n\}$ be a bounded sequence. Its limit superior and limit inferior are

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} \big(\sup\{x_k : k \geq n\}\big), \qquad \liminf_{n\to\infty} x_n := \lim_{n\to\infty} \big(\inf\{x_k : k \geq n\}\big).$$

We often write $\limsup x_n$ and $\liminf x_n$.

Theorem 9.3. *Let $\{x_n\}$ be bounded, and set $a_n = \sup\{x_k : k \geq n\}$ and $b_n = \inf\{x_k : k \geq n\}$. Then*

1. *$\{a_n\}$ is monotone decreasing and bounded, and $\{b_n\}$ is monotone increasing and bounded; hence both limits in the definition exist;*
2. *$\liminf x_n \leq \limsup x_n$.*

Proof. (1) Since $\{x_k : k \geq n+1\} \subseteq \{x_k : k \geq n\}$, the supremum of the smaller set is at most that of the larger, so $a_{n+1} \leq a_n$; similarly $b_{n+1} \geq b_n$. If $-B \leq x_n \leq B$ for all n, then $-B \leq b_n \leq a_n \leq B$ for all n. The monotone convergence theorem (Lecture 7) gives the limits.
(2) Since $b_n \leq a_n$ for all n, the order property of limits (Lecture 8) gives $\lim b_n \leq \lim a_n$. □

Example: *Two computations.* For $x_n = (-1)^n$, every tail $\{(-1)^k : k \geq n\}$ equals $\{-1, 1\}$, so $a_n = 1$ and $b_n = -1$ for all n. Thus $\limsup x_n = 1$ and $\liminf x_n = -1$. For $x_n = 1/n$, $a_n = \sup\{1/k : k \geq n\} = 1/n \to 0$ and $b_n = \inf\{1/k : k \geq n\} = 0$, so $\limsup x_n = \liminf x_n = 0$.

9.3 The Bolzano-Weierstrass theorem

Theorem 9.4. *Let $\{x_n\}$ be bounded. There are subsequences $\{x_{n_k}\}$ and $\{x_{m_k}\}$ with*

$$\lim_{k\to\infty} x_{n_k} = \limsup_{n\to\infty} x_n, \qquad \lim_{k\to\infty} x_{m_k} = \liminf_{n\to\infty} x_n.$$

Proof. Let $a_n = \sup\{x_k : k \geq n\}$. Since $a_1 - 1$ is not an upper bound of $\{x_k : k \geq 1\}$, there is n_1 with $a_1 - 1 < x_{n_1} \leq a_1$. Inductively, given n_k, the number $a_{n_k+1} - \frac{1}{k+1}$ is not an upper bound of $\{x_j : j \geq n_k + 1\}$, so there is $n_{k+1} \geq n_k + 1$ with

$$a_{n_k+1} - \frac{1}{k+1} < x_{n_{k+1}} \leq a_{n_k+1}.$$

This gives $n_1 < n_2 < \cdots$, a subsequence. The sequence $\{a_{n_k+1}\}_k$ is a subsequence of $\{a_n\}$, so it converges to $\limsup x_n$ (Lecture 7), and so does $a_{n_k+1} - \frac{1}{k+1}$. By the squeeze theorem, $x_{n_{k+1}} \to \limsup x_n$, so $x_{n_k} \to \limsup x_n$.
For the limit inferior, choose instead $b_{m_k+1} \leq x_{m_{k+1}} < b_{m_k+1} + \frac{1}{k+1}$, where $b_n = \inf\{x_k : k \geq n\}$, and argue the same way. □

Theorem 9.5 [Bolzano-Weierstrass]. *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. Take the subsequence converging to $\limsup x_n$ from Theorem 9.4. □

Theorem 9.6. *A bounded sequence $\{x_n\}$ converges if and only if $\liminf x_n = \limsup x_n$. In that case, $\lim x_n = \liminf x_n = \limsup x_n$.*

Proof. ($\Leftarrow$) For every n, $\inf\{x_k : k \geq n\} \leq x_n \leq \sup\{x_k : k \geq n\}$. The outer sequences have the same limit by assumption, so the squeeze theorem gives $\lim x_n = \liminf x_n = \limsup x_n$.
($\Rightarrow$) Let $x_n \to x$. Every subsequence converges to x (Lecture 7). The subsequences of Theorem 9.4 converge to $\limsup x_n$ and to $\liminf x_n$, so both equal x by uniqueness of limits. □

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Lecture 10: The Completeness of the Real Numbers and Basic Properties of Infinite Series

10.1 Cauchy sequences

Definition 10.1 [Cauchy sequence]. A sequence $\{x_n\}$ is Cauchy if for every $\epsilon > 0$ there is $M \in \mathbb{N}$ such that

$$|x_n - x_k| < \epsilon \qquad \text{for all } n, k \geq M.$$

The terms get close to *each other*, with no candidate limit mentioned. Negating, $\{x_n\}$ is not Cauchy if there is $\epsilon_0 > 0$ such that for every $M \in \mathbb{N}$ some $n, k \geq M$ have $|x_n - x_k| \geq \epsilon_0$.

Example: *$1/n$ is Cauchy; $(-1)^n$ is not.* Given $\epsilon > 0$, choose M with $1/M < \epsilon/2$. For $n, k \geq M$,

$$\left|\frac{1}{n} - \frac{1}{k}\right| \leq \frac{1}{n} + \frac{1}{k} \leq \frac{2}{M} < \epsilon.$$

For $x_n = (-1)^n$, take $\epsilon_0 = 1$: for every M, the indices $n = M$ and $k = M+1$ give $|(-1)^n - (-1)^k| = 2 \geq 1$.

Theorem 10.2. *Every Cauchy sequence is bounded.*

Proof. Choose M with $|x_n - x_k| < 1$ for $n, k \geq M$. For $n \geq M$, $|x_n| \leq |x_n - x_M| + |x_M| < |x_M| + 1$. Hence $B = |x_1| + \cdots + |x_M| + 1$ bounds every $|x_n|$. □

Theorem 10.3. *If $\{x_n\}$ is Cauchy and some subsequence $\{x_{n_k}\}$ converges to x, then $x_n \to x$.*

Proof. Let $\epsilon > 0$. Choose M_0 with $|x_{n_k} - x| < \epsilon/2$ for $k \geq M_0$, and M_1 with $|x_n - x_m| < \epsilon/2$ for $n, m \geq M_1$. Let $M = M_0 + M_1$ and $n \geq M$. The index $k = M$ satisfies $k \geq M_0$, and $n_M \geq M \geq M_1$ (since $n_k \geq k$). Therefore

$$|x_n - x| \leq |x_n - x_{n_M}| + |x_{n_M} - x| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon.$$ □

Theorem 10.4 [Completeness of $\mathbb{R}$]. *A sequence of real numbers is Cauchy if and only if it is convergent.*

Proof. ($\Rightarrow$) A Cauchy sequence is bounded by Theorem 10.2, so it has a convergent subsequence by the Bolzano-Weierstrass theorem (Lecture 9). It converges by Theorem 10.3.
($\Leftarrow$) Let $x_n \to x$ and $\epsilon > 0$. Choose M with $|x_n - x| < \epsilon/2$ for $n \geq M$. For $n, k \geq M$, $|x_n - x_k| \leq |x_n - x| + |x - x_k| < \epsilon$. □

10.2 Infinite series

Series were the original motivation for analysis.

Definition 10.5 [Series]. Given a sequence $\{x_n\}$, the symbol $\sum_{n=1}^{\infty} x_n$ (or $\sum x_n$) is the associated series. Its partial sums are $s_m = \sum_{n=1}^{m} x_n$. The series converges if $\{s_m\}_{m=1}^{\infty}$ converges; if $s_m \to s$, we write $s = \sum_{n=1}^{\infty} x_n$ and treat the series as this number.

A series need not start at $n = 1$; the same definition applies to $\sum_{n=0}^{\infty} x_n$ or $\sum_{n=M}^{\infty} x_n$.

Example: *A telescoping series.* Since $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, the partial sums telescope:

$$s_m = \sum_{n=1}^{m} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \left(1 + \frac{1}{2} + \cdots + \frac{1}{m} \right) - \left(\frac{1}{2} + \cdots + \frac{1}{m} + \frac{1}{m+1} \right) = 1 - \frac{1}{m+1}.$$

So $s_m \to 1$ and $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$.

Theorem 10.6 [Geometric series]. *If $|r| < 1$, then $\sum_{n=0}^{\infty} r^n$ converges and*

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Proof. By induction (Lecture 1), $s_m = \sum_{n=0}^{m} r^n = \frac{1-r^{m+1}}{1-r}$. Since $|r| < 1$, $|r^{m+1}| = |r|^{m+1} \to 0$ (Lecture 8), so $s_m \to \frac{1-0}{1-r} = \frac{1}{1-r}$. □

A series $\sum_{n=0}^{\infty} \alpha r^n$ with $\alpha, r \in \mathbb{R}$ is called a geometric series.

Theorem 10.7 [Tails]. *Let $M \in \mathbb{N}$. Then $\sum_{n=1}^{\infty} x_n$ converges if and only if $\sum_{n=M}^{\infty} x_n$ converges.*

Proof. For $m \geq M$, the partial sums differ by a constant:

$$\sum_{n=1}^{m} x_n = \sum_{n=M}^{m} x_n + \sum_{n=1}^{M-1} x_n.$$

So one sequence of partial sums converges if and only if the other does. □

10.3 The Cauchy criterion for series

Definition 10.8 [Cauchy series]. A series $\sum x_n$ is Cauchy if its sequence of partial sums is Cauchy.

By Theorem 10.4, a series is Cauchy if and only if it converges.

Theorem 10.9 [Cauchy criterion]. *$\sum x_n$ is Cauchy if and only if for every $\epsilon > 0$ there is $M \in \mathbb{N}$ such that*

$$\left| \sum_{n=m+1}^{\ell} x_n \right| < \epsilon \qquad \text{for all } \ell > m \geq M.$$

Proof. For $\ell > m$, $\sum_{n=m+1}^{\ell} x_n = s_\ell - s_m$.
($\Rightarrow$) Choose M with $|s_m - s_\ell| < \epsilon$ for $m, \ell \geq M$; in particular for $\ell > m \geq M$.
($\Leftarrow$) Let $\epsilon > 0$ and take M from the hypothesis. For $m, \ell \geq M$: if $\ell > m$ then $|s_\ell - s_m| < \epsilon$ by hypothesis; if $m > \ell$, swap the roles; if $m = \ell$, the difference is 0. So $\{s_m\}$ is Cauchy. □

Theorem 10.10 [Term test]. *If $\sum x_n$ converges, then $\lim_{n\to\infty} x_n = 0$.*

Proof. The series is Cauchy. Given $\epsilon > 0$, choose M_0 as in Theorem 10.9. If $m \geq M_0 + 1$, then $m - 1 \geq M_0$, and the criterion with the pair $(m-1, m)$ gives

$$|x_m| = \left| \sum_{n=m}^{m} x_n \right| < \epsilon.$$ □

Theorem 10.11. *If $|r| \geq 1$, then $\sum_{n=0}^{\infty} r^n$ diverges.*

Proof. Here $|r^m| = |r|^m \geq 1$ for all m, so $r^m \not\to 0$. By the term test, the series cannot converge. □

Corollary 10.12. *For $\alpha \neq 0$, the geometric series $\sum_{n=0}^{\infty} \alpha r^n$ converges if and only if $|r| < 1$.*

Proof. If $|r| < 1$, the partial sums are α times those of Theorem 10.6, so the series converges to $\alpha/(1-r)$. If $|r| \geq 1$, then $|\alpha r^m| \geq |\alpha| > 0$, so the terms do not tend to 0 and the series diverges. □

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Lecture 11: Absolute Convergence and the Comparison Test for Series

Throughout, $s_m = \sum_{n=1}^{m} x_n$ denotes the mth partial sum of the series $\sum x_n$. If $\sum x_n$ converges then $x_n \to 0$. This unit shows that the converse fails, develops the basic tools for series with nonnegative terms, and decides exactly which p-series converge.

11.1 The harmonic series

Theorem 11.1 [Harmonic series]. *The series $\sum_{n=1}^{\infty} \frac{1}{n}$, called the* harmonic series, *diverges, even though $\frac{1}{n} \to 0$.*

Proof. It suffices to find an unbounded subsequence of the partial sums, since a convergent sequence is bounded. Group the terms of s_{2^ℓ} into blocks $2^{\lambda-1} < n \le 2^\lambda$:

$$s_{2^\ell} = 1 + \left(\tfrac{1}{2}\right) + \left(\tfrac{1}{3} + \tfrac{1}{4}\right) + \left(\tfrac{1}{5} + \cdots + \tfrac{1}{8}\right) + \cdots + \left(\tfrac{1}{2^{\ell-1}+1} + \cdots + \tfrac{1}{2^\ell}\right).$$

Block λ has $2^\lambda - 2^{\lambda-1} = 2^{\lambda-1}$ terms, each at least $2^{-\lambda}$, so it contributes at least $\frac{1}{2}$:

$$s_{2^\ell} = 1 + \sum_{\lambda=1}^{\ell} \sum_{n=2^{\lambda-1}+1}^{2^\lambda} \frac{1}{n} \;\ge\; 1 + \sum_{\lambda=1}^{\ell} \frac{2^{\lambda-1}}{2^\lambda} = 1 + \frac{\ell}{2}.$$

Thus $\{s_{2^\ell}\}_{\ell=1}^{\infty}$ is unbounded, and $\{s_m\}$ does not converge. □

11.2 Linearity and series with nonnegative terms

Theorem 11.2 [Linearity]. *If $\alpha \in \mathbb{R}$ and $\sum x_n$, $\sum y_n$ converge, then $\sum(\alpha x_n + y_n)$ converges and*

$$\sum(\alpha x_n + y_n) = \alpha \sum x_n + \sum y_n.$$

Proof. The partial sums satisfy $\sum_{n=1}^{m}(\alpha x_n + y_n) = \alpha \sum_{n=1}^{m} x_n + \sum_{n=1}^{m} y_n$; let $m \to \infty$ and use the limit laws for sequences. □

Theorem 11.3 [Nonnegative terms]. *If $x_n \ge 0$ for all n, then $\sum x_n$ converges if and only if the partial sums $\{s_m\}$ are bounded.*

Proof. Since $s_{m+1} = s_m + x_{m+1} \geq s_m$, the sequence $\{s_m\}$ is monotone increasing, and a monotone sequence converges if and only if it is bounded. □

11.3 Absolute convergence

Definition 11.4 [Absolute convergence]. The series $\sum x_n$ converges absolutely if $\sum |x_n|$ converges.

Theorem 11.5 [Absolute convergence implies convergence]. *If $\sum x_n$ converges absolutely, then $\sum x_n$ converges.*

Proof. We show that $\sum x_n$ is Cauchy. First, the triangle inequality extends to finite sums: $\left|\sum_{n=1}^{m} x_n\right| \leq \sum_{n=1}^{m} |x_n|$ for all m. The case $m = 2$ is the triangle inequality, and the inductive step is

$$\left|\sum_{n=1}^{\ell+1} x_n\right| \leq \left|\sum_{n=1}^{\ell} x_n\right| + |x_{\ell+1}| \leq \sum_{n=1}^{\ell} |x_n| + |x_{\ell+1}| = \sum_{n=1}^{\ell+1} |x_n|.$$

Let $\epsilon > 0$. Since $\sum |x_n|$ converges, it is Cauchy: there is $M \in \mathbb{N}$ such that $\sum_{n=m+1}^{\ell} |x_n| < \epsilon$ for all $\ell > m \geq M$. For the same ℓ, m,

$$\left|\sum_{n=m+1}^{\ell} x_n\right| \leq \sum_{n=m+1}^{\ell} |x_n| < \epsilon.$$

So $\sum x_n$ is Cauchy, hence convergent. □

The converse is false. The series $\sum \frac{(-1)^n}{n}$ converges (this follows from the alternating series test of Lecture 12), but $\sum \left|\frac{(-1)^n}{n}\right| = \sum \frac{1}{n}$ is the harmonic series, which diverges.

11.4 The comparison test and p-series

Theorem 11.6 [Comparison test]. *Suppose $0 \leq x_n \leq y_n$ for all $n \in \mathbb{N}$.*

1. *If $\sum y_n$ converges, then $\sum x_n$ converges.*
2. *If $\sum x_n$ diverges, then $\sum y_n$ diverges.*

Proof. Both statements rest on Theorem 11.3 and the termwise bound $\sum_{n=1}^{m} x_n \leq \sum_{n=1}^{m} y_n$.

1. If $\sum y_n$ converges, there is $B \geq 0$ with $\sum_{n=1}^{m} y_n \leq B$ for all m. Then $\sum_{n=1}^{m} x_n \leq B$ for all m, so the partial sums of $\sum x_n$ are bounded and $\sum x_n$ converges.
2. If $\sum x_n$ diverges, its partial sums are unbounded: for each $B \geq 0$ some m has $\sum_{n=1}^{m} x_n \geq B$,

and then $\sum_{n=1}^{m} y_n \geq B$. So the partial sums of $\sum y_n$ are unbounded and $\sum y_n$ diverges. □

Geometric series and the comparison test together drive nearly all the convergence tests that follow. The first instance is the p-series.

Theorem 11.7 [p-series]. *For $p \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.*

Proof. ($\Rightarrow$) Suppose $p \leq 1$. Then $\frac{1}{n^p} \geq \frac{1}{n}$ for all n, and $\sum \frac{1}{n}$ diverges, so $\sum \frac{1}{n^p}$ diverges by the comparison test. Hence convergence forces $p > 1$.

($\Leftarrow$) Suppose $p > 1$, and let $s_m = \sum_{n=1}^{m} n^{-p}$. Group the terms of s_{2^k} into the same blocks as for the harmonic series. Block ℓ has $2^{\ell-1}$ terms, each at most $(2^{\ell-1}+1)^{-p} \leq 2^{-p(\ell-1)}$, so

$$s_{2^k} = 1 + \sum_{\ell=1}^{k} \sum_{n=2^{\ell-1}+1}^{2^\ell} \frac{1}{n^p} \leq 1 + \sum_{\ell=1}^{k} 2^{-p(\ell-1)}\, 2^{\ell-1} = 1 + \sum_{\ell=0}^{k-1} 2^{-(p-1)\ell} \leq 1 + \frac{1}{1-2^{-(p-1)}},$$

summing the geometric series with ratio $2^{-(p-1)} < 1$ (here $p - 1 > 0$ is used). For any $m \in \mathbb{N}$, since $2^m > m$ and the terms are positive,

$$s_m \leq s_{2^m} \leq 1 + \frac{1}{1-2^{-(p-1)}}.$$

So the partial sums are bounded and the series converges by Theorem 11.3. □

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Lecture 12: The Ratio, Root, and Alternating Series Tests

The ratio and root tests compare a series with a geometric series; the alternating series test handles some series that converge but not absolutely. Absolute convergence also makes the value of a series independent of the order of its terms.

12.1 The ratio test

Theorem 12.1 [Ratio test]. *Suppose $x_n \neq 0$ for all n and the limit*

$$L = \lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|}$$

exists.

1. *If $L < 1$, then $\sum x_n$ converges absolutely.*
2. *If $L > 1$, then $\sum x_n$ diverges.*

Proof. *Case $L > 1$.* Choose $\alpha \in (1, L)$. There is M_0 such that $|x_{n+1}|/|x_n| \geq \alpha > 1$ for all $n \geq M_0$. Then $|x_{n+1}| \geq |x_n|$ for $n \geq M_0$, so $|x_n| \geq |x_{M_0}| > 0$ for all $n \geq M_0$. Hence $x_n \not\to 0$ and $\sum x_n$ diverges.

Case $L < 1$. Choose $\alpha \in (L, 1)$. There is M_0 such that $|x_{n+1}| \leq \alpha |x_n|$ for all $n \geq M_0$, and iterating gives

$$|x_n| \leq \alpha|x_{n-1}| \leq \alpha^2|x_{n-2}| \leq \cdots \leq \alpha^{n-M_0}|x_{M_0}| \qquad (n \geq M_0).$$

Comparing the tail with a geometric series, for every m,

$$\sum_{n=1}^{m} |x_n| \leq \sum_{n=1}^{M_0-1} |x_n| + |x_{M_0}| \sum_{n=M_0}^{m} \alpha^{n-M_0} \leq \sum_{n=1}^{M_0-1} |x_n| + \frac{|x_{M_0}|}{1-\alpha}.$$

The partial sums of $\sum |x_n|$ are bounded, so $\sum x_n$ converges absolutely. □

If $L = 1$, the test says nothing.

Example: *The exponential series.* For every $x \in \mathbb{R}$, the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges absolutely. For $x \neq 0$,

$$\lim_{n\to\infty} \frac{|x|^{n+1}}{(n+1)!} \cdot \frac{n!}{|x|^n} = \lim_{n\to\infty} \frac{|x|}{n+1} = 0 < 1,$$

and for $x = 0$ only the first term is nonzero.

Example: *An inconclusive ratio.* The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2+1}$ converges absolutely, because $\left|\frac{(-1)^n}{n^2+1}\right| \le \frac{1}{n^2}$ and $\sum \frac{1}{n^2}$ converges (p-series with $p = 2$, Lecture 11) by the comparison test. The ratio test cannot show this: here

$$\frac{|x_{n+1}|}{|x_n|} = \frac{n^2+1}{(n+1)^2+1} \longrightarrow 1.$$

12.2 The root test

Theorem 12.2 [Root test]. *Suppose the limit $L = \lim_{n\to\infty} |x_n|^{1/n}$ exists.*

1. *If $L < 1$, then $\sum x_n$ converges absolutely.*
2. *If $L > 1$, then $\sum x_n$ diverges.*

Proof. *Case $L < 1$.* Choose $r \in (L, 1)$. There is M with $|x_n|^{1/n} < r$, that is $|x_n| < r^n$, for all $n \ge M$. For every m,

$$\sum_{n=1}^{m} |x_n| \le \sum_{n=1}^{M-1} |x_n| + \sum_{n=M}^{\infty} r^n = \sum_{n=1}^{M-1} |x_n| + \frac{r^M}{1-r},$$

so the partial sums of $\sum |x_n|$ are bounded.

Case $L > 1$. There is M with $|x_n|^{1/n} > 1$, so $|x_n| > 1$, for all $n \ge M$. Then $x_n \not\to 0$ and $\sum x_n$ diverges. □

Again the test says nothing when $L = 1$.

12.3 The alternating series test

Theorem 12.3 [Alternating series test]. *If $\{x_n\}$ is monotone decreasing and $x_n \to 0$, then $\sum (-1)^n x_n$ converges.*

Proof. Let $s_m = \sum_{n=1}^{m} (-1)^n x_n$. Note $x_n \ge 0$, since a decreasing sequence lies above its limit. Pairing consecutive terms,

$$s_{2k} = (x_2 - x_1) + (x_4 - x_3) + \cdots + (x_{2k} - x_{2k-1}),$$

and $s_{2k+2} = s_{2k} + (x_{2k+2} - x_{2k+1}) \le s_{2k}$, so $\{s_{2k}\}$ is monotone decreasing. Pairing the other way,

$$s_{2k} = -x_1 + (x_2 - x_3) + (x_4 - x_5) + \cdots + (x_{2k-2} - x_{2k-1}) + x_{2k} \ge -x_1.$$

A decreasing sequence bounded below converges; let $s = \lim_k s_{2k}$.

It remains to show $s_m \to s$, including odd m. Let $\epsilon > 0$. Choose M_0 with $|s_{2k} - s| < \epsilon/2$ for

$k \geq M_0$, and M_1 with $|x_n| < \epsilon/2$ for $n \geq M_1$. Let $M = \max\{2M_0 + 1, M_1\}$ and $m \geq M$. If m is even, then $m/2 \geq M_0$ and $|s_m - s| < \epsilon/2$. If $m = 2k + 1$ is odd, then $k \geq M_0$ and $m \geq M_1$, so

$$|s_m - s| = |s_{2k} - x_m - s| \leq |s_{2k} - s| + |x_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \qquad \square$$

Corollary 12.4 [Alternating harmonic series]. *The series $\sum \frac{(-1)^n}{n}$ converges, although it does not converge absolutely (its absolute series is the harmonic series).*

12.4 Rearrangements of absolutely convergent series

Theorem 12.5 [Rearrangement]. *Suppose $\sum x_n$ converges absolutely with $\sum x_n = x$, and let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection. Then $\sum x_{\sigma(n)}$ converges absolutely and $\sum x_{\sigma(n)} = x$.*

Proof. Absolute convergence. There is $B \geq 0$ with $\sum_{n=1}^{\ell} |x_n| \leq B$ for all ℓ. Given m, the finite set $\sigma(\{1, \ldots, m\})$ lies in some $\{1, \ldots, \ell\}$, and σ is injective, so

$$\sum_{n=1}^{m} |x_{\sigma(n)}| = \sum_{n \in \sigma(\{1,\ldots,m\})} |x_n| \leq \sum_{n=1}^{\ell} |x_n| \leq B.$$

Bounded partial sums of nonnegative terms give convergence of $\sum |x_{\sigma(n)}|$.

Same sum. Let $\epsilon > 0$. Choose M_2 such that $\left|\sum_{n=1}^{M_2} x_n - x\right| < \epsilon/2$ and $\sum_{n=M_2+1}^{\ell} |x_n| < \epsilon/2$ for all $\ell > M_2$; this is possible because $\sum x_n$ converges and $\sum |x_n|$ is Cauchy. Since σ is surjective, the finite set $\{1, \ldots, M_2\}$ lies in $\sigma(\{1, \ldots, M_3\})$ for some M_3. For $m' \geq M_3$, the set $\sigma(\{1, \ldots, m'\})$ contains $\{1, \ldots, M_2\}$; let ℓ be its largest element. Then

$$\left|\sum_{n'=1}^{m'} x_{\sigma(n')} - x\right| = \left|\sum_{n=1}^{M_2} x_n - x + \sum_{n \in \sigma(\{1,\ldots,m'\}) \setminus \{1,\ldots,M_2\}} x_n\right| \leq \left|\sum_{n=1}^{M_2} x_n - x\right| + \sum_{n=M_2+1}^{\ell} |x_n| < \epsilon,$$

where the last sum is empty if $\ell = M_2$. $\square$

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 12, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 13: Limits of Functions

Informally, a function is continuous when sufficiently small changes in the input produce tolerably small changes in the output. Making this precise starts with limits of functions, which require the point being approached to be a limit of points of the domain.

13.1 Cluster points

Definition 13.1 [Cluster point]. Let $S \subset \mathbb{R}$. A point $x \in \mathbb{R}$ is a cluster point of S if for every $\delta > 0$,

$$(x - \delta, x + \delta) \cap S \setminus \{x\} \neq \emptyset.$$

A cluster point need not belong to S, and a point of S need not be a cluster point.

- 0 is a cluster point of $S = \{1/n : n \in \mathbb{N}\}$.
- The set of cluster points of $(0, 1)$ is $[0, 1]$.
- The set of cluster points of $\mathbb{Q}$ is $\mathbb{R}$.
- $\{0\}$ and $\mathbb{Z}$ have no cluster points.

Theorem 13.2 [Sequential characterization of cluster points]. *Let $S \subset \mathbb{R}$. Then x is a cluster point of S if and only if there is a sequence $\{x_n\}$ of elements of $S \setminus \{x\}$ with $x_n \to x$.*

Proof. If x is a cluster point, pick $x_n \in (x - \frac{1}{n}, x + \frac{1}{n}) \cap S \setminus \{x\}$ for each n; then $|x_n - x| < \frac{1}{n}$, so $x_n \to x$. Conversely, given such a sequence and $\delta > 0$, some x_n satisfies $|x_n - x| < \delta$, and it lies in $(x - \delta, x + \delta) \cap S \setminus \{x\}$. □

13.2 Limits of functions

Definition 13.3 [Limit of a function]. Let $S \subset \mathbb{R}$, let c be a cluster point of S, and let $f : S \to \mathbb{R}$. Then $f(x)$ converges to $L \in \mathbb{R}$ as $x \to c$ if for every $\epsilon > 0$ there is $\delta > 0$ such that

$$x \in S \text{ and } 0 < |x - c| < \delta \quad \Longrightarrow \quad |f(x) - L| < \epsilon.$$

We write $f(x) \to L$ as $x \to c$, or $\lim_{x \to c} f(x) = L$.

The condition $0 < |x - c|$ means the value $f(c)$, if defined, plays no role. Requiring c to be a cluster

point guarantees that points x with $0 < |x - c| < \delta$ exist, which is what makes the limit unique.

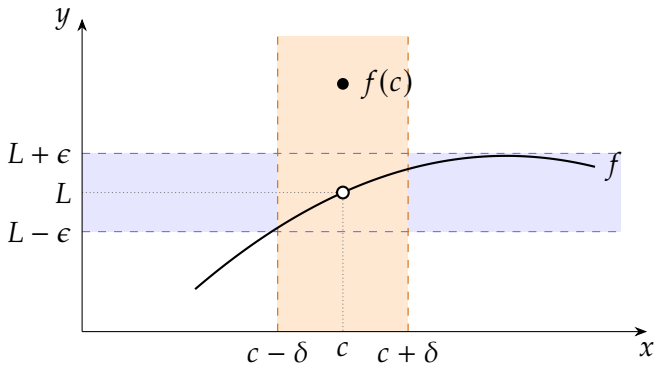


Figure 5: Schematic of $\lim_{x \to c} f(x) = L$. Every $x \neq c$ in the window $(c - \delta, c + \delta)$ has $f(x)$ in the band $(L - \epsilon, L + \epsilon)$. The value $f(c)$ (filled dot) is irrelevant.

Theorem 13.4 [Uniqueness of limits]. *Let c be a cluster point of S and $f : S \to \mathbb{R}$. If $f(x) \to L_1$ and $f(x) \to L_2$ as $x \to c$, then $L_1 = L_2$.*

Proof. We show $|L_1 - L_2| < \epsilon$ for every $\epsilon > 0$. Choose δ_1 and δ_2 so that $0 < |x - c| < \delta_i$ and $x \in S$ imply $|f(x) - L_i| < \epsilon/2$, and let $\delta = \min\{\delta_1, \delta_2\}$. Because c is a cluster point, there is $x_0 \in S$ with $0 < |x_0 - c| < \delta$. Then

$$|L_1 - L_2| \leq |L_1 - f(x_0)| + |f(x_0) - L_2| < \epsilon. \qquad \square$$

Example: *Linear functions.* For $f(x) = ax + b$ and every $c \in \mathbb{R}$, $\lim_{x \to c} f(x) = ac + b$. Given $\epsilon > 0$, take $\delta = \frac{\epsilon}{1 + |a|}$; then $0 < |x - c| < \delta$ gives

$$|f(x) - (ac + b)| = |a||x - c| < |a|\delta = \frac{|a|}{1 + |a|}\,\epsilon < \epsilon.$$

The $1 + |a|$ avoids dividing by zero when $a = 0$.

Example: *Square root.* For $f(x) = \sqrt{x}$ on $[0, \infty)$ and every $c > 0$, $\lim_{x \to c} \sqrt{x} = \sqrt{c}$. Given $\epsilon > 0$, take $\delta = \epsilon\sqrt{c}$. If $x \geq 0$ and $0 < |x - c| < \delta$, rationalizing gives

$$|\sqrt{x} - \sqrt{c}| = \frac{|x - c|}{\sqrt{x} + \sqrt{c}} \leq \frac{|x - c|}{\sqrt{c}} < \frac{\delta}{\sqrt{c}} = \epsilon.$$

Example: *The limit ignores $f(c)$.* Let $f(x) = 1$ for $x \neq 0$ and $f(0) = 2$. Then $\lim_{x \to 0} f(x) = 1 \neq f(0)$: with $\delta = 1$, every x with $0 < |x| < 1$ has $|f(x) - 1| = 0 < \epsilon$.

13.3 Limits of functions through sequences

Limits of functions reduce to limits of sequences, so earlier sequence results transfer directly.

Theorem 13.5 [Sequential criterion for limits]. *Let $S \subset \mathbb{R}$, let c be a cluster point of S, and let $f : S \to \mathbb{R}$. The following are equivalent:*

1. $\lim_{x \to c} f(x) = L$;
2. *for every sequence $\{x_n\}$ in $S \setminus \{c\}$ with $x_n \to c$, we have $f(x_n) \to L$.*

Proof. $(1 \Rightarrow 2)$ Let $x_n \to c$ with $x_n \in S \setminus \{c\}$, and let $\epsilon > 0$. Take δ from the definition of the limit. Since $x_n \to c$ and $x_n \neq c$, there is M with $0 < |x_n - c| < \delta$ for all $n \geq M$, hence $|f(x_n) - L| < \epsilon$ for all $n \geq M$.

$(2 \Rightarrow 1)$ Suppose 2 holds but 1 fails. Then there is $\epsilon_0 > 0$ such that for every $\delta > 0$ some $x \in S$ has $0 < |x - c| < \delta$ and $|f(x) - L| \geq \epsilon_0$. Taking $\delta = \frac{1}{n}$ gives $x_n \in S$ with

$$0 < |x_n - c| < \frac{1}{n} \qquad \text{and} \qquad |f(x_n) - L| \geq \epsilon_0.$$

Then $x_n \to c$ by the squeeze theorem and $x_n \neq c$, so 2 gives the contradiction

$$0 = \lim_{n \to \infty} |f(x_n) - L| \geq \epsilon_0. \qquad \square$$

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 13, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 14: Limits of Functions in Terms of Sequences and Continuity

The sequential criterion of Lecture 13 turns questions about limits of functions into questions about limits of sequences. This unit uses it for examples and limit inequalities, introduces one-sided limits, and defines continuity.

14.1 Computing limits with sequences

Example: *Squares.* For all $c \in \mathbb{R}$, $\lim_{x\to c} x^2 = c^2$. If $x_n \to c$ with $x_n \neq c$, then $x_n^2 \to c^2$ because limits of sequences respect products (Lecture 8); the sequential criterion gives the claim.

Theorem 14.1 [Two oscillating examples]**.**

1. $\lim_{x\to 0} \sin(1/x)$ *does not exist.*
2. $\lim_{x\to 0} x\sin(1/x) = 0$.

Proof. 1. Let $x_n = \frac{2}{(2n-1)\pi}$. Then $x_n \neq 0$ and $x_n \to 0$, but

$$\sin(1/x_n) = \sin\left(\frac{(2n-1)\pi}{2}\right) = (-1)^{n+1},$$

which does not converge. By the sequential criterion, no limit L exists.

2. If $x_n \neq 0$ and $x_n \to 0$, then $0 \leq |x_n \sin(1/x_n)| \leq |x_n|$, so $x_n \sin(1/x_n) \to 0$ by the squeeze theorem. □

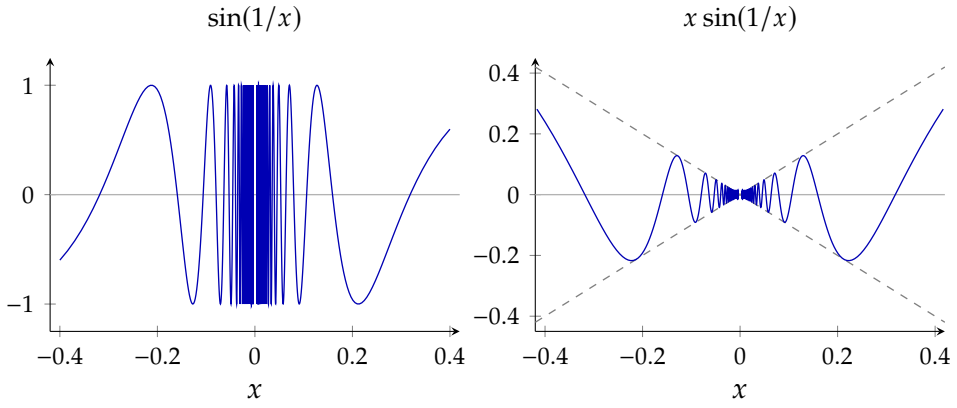


Figure 6: Left: $\sin(1/x)$ takes every value in $[-1, 1]$ on each interval $(0, \delta)$, so it has no limit at 0. Right: $x\sin(1/x)$ is trapped between $-|x|$ and $|x|$ (dashed), so it tends to 0.

The sequential criterion transfers sequence results to function limits: the squeeze theorem, limits of sums, products, quotients and absolute values all have analogs for $\lim_{x\to c}$. For instance:

Theorem 14.2 [Limits preserve inequalities]. *Let c be a cluster point of S and $f, g : S \to \mathbb{R}$ with $f(x) \le g(x)$ for all $x \in S$. If $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ exist, then*

$$\lim_{x\to c} f(x) \le \lim_{x\to c} g(x).$$

Proof. Let L_1, L_2 be the two limits and take any sequence $\{x_n\}$ in $S \setminus \{c\}$ with $x_n \to c$ (one exists by Lecture 13). Then $f(x_n) \le g(x_n)$ for all n, so $L_1 = \lim f(x_n) \le \lim g(x_n) = L_2$. □

14.2 One-sided limits

Definition 14.3 [One-sided limits]. Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$. If c is a cluster point of $S \cap (-\infty, c)$, then $f(x) \to L$ as $x \to c^-$, written $L = \lim_{x\to c^-} f(x)$, if for every $\epsilon > 0$ there is $\delta > 0$ such that $x \in S$ and $c - \delta < x < c$ imply $|f(x) - L| < \epsilon$. Similarly, if c is a cluster point of $S \cap (c, \infty)$, then $L = \lim_{x\to c^+} f(x)$ if for every $\epsilon > 0$ there is $\delta > 0$ such that $x \in S$ and $c < x < c + \delta$ imply $|f(x) - L| < \epsilon$.

Example: Let $f(x) = 1$ for $x > 0$ and $f(x) = 0$ for $x < 0$, with $f(0)$ undefined. Then $\lim_{x\to 0^-} f(x) = 0$ and $\lim_{x\to 0^+} f(x) = 1$, so $\lim_{x\to 0} f(x)$ does not exist by the next theorem.

Theorem 14.4 [Limits from one-sided limits]. *Let c be a cluster point of both $S \cap (-\infty, c)$ and $S \cap (c, \infty)$. Then c is a cluster point of S, and*

$$\lim_{x\to c} f(x) = L \quad \Longleftrightarrow \quad \lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = L.$$

Proof. Points of $S \cap (-\infty, c)$ near c are points of $S \setminus \{c\}$ near c, so c is a cluster point of S.

($\Rightarrow$) A δ that works for the two-sided limit works for each side.

($\Leftarrow$) Given ϵ, take δ_- and δ_+ for the two sides and $\delta = \min\{\delta_-, \delta_+\}$. Any $x \in S$ with $0 < |x - c| < \delta$ lies on one side within the corresponding $\delta_\pm$. □

14.3 Continuity

Limits ignore $f(c)$; continuity is the condition that ties the behavior near c to the value at c.

Definition 14.5 [Continuity]. Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \to \mathbb{R}$. Then f is continuous at c if for

every $\epsilon > 0$ there is $\delta > 0$ such that

$$x \in S \text{ and } |x - c| < \delta \quad \Longrightarrow \quad |f(x) - f(c)| < \epsilon.$$

For $U \subset S$, f is CONTINUOUS ON U if it is continuous at every point of U.

Negating the definition: f is *not* continuous at c if there is $\epsilon_0 > 0$ such that for every $\delta > 0$ some $x \in S$ has $|x - c| < \delta$ and $|f(x) - f(c)| \geq \epsilon_0$.

Example: *Linear functions are continuous on* $\mathbb{R}$. For $f(x) = ax + b$ and $c \in \mathbb{R}$, the choice $\delta = \frac{\epsilon}{1+|a|}$ from the limit example in Lecture 13 gives $|f(x) - f(c)| = |a||x - c| < \epsilon$ whenever $|x - c| < \delta$.

Example: *Changing one value.* Let $f(x) = 1$ for $x \neq 0$ and $f(0) = 2$. Then f is not continuous at 0: take $\epsilon_0 = 1$. For any $\delta > 0$, the point $x = \delta/2$ has $|x - 0| < \delta$ and $|f(x) - f(0)| = |1 - 2| = 1 \geq \epsilon_0$.

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 14, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 15: The Continuity of Sine and Cosine and the Many Discontinuities of Dirichlet's Function

Continuity, like limits, can be tested with sequences. This gives continuity of sine, cosine, polynomials, and their algebraic combinations and compositions with little ϵ-δ work, while Dirichlet's function is continuous nowhere.

15.1 Continuity via limits and sequences

Theorem 15.1 [Characterizations of continuity]. *Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \to \mathbb{R}$.*

1. *If c is not a cluster point of S, then f is continuous at c.*
2. *If c is a cluster point of S, then f is continuous at c if and only if $\lim_{x\to c} f(x) = f(c)$.*
3. *f is continuous at c if and only if $f(x_n) \to f(c)$ for every sequence $\{x_n\}$ in S with $x_n \to c$.*

Proof. 1. There is $\delta_0 > 0$ with $(c-\delta_0, c+\delta_0) \cap S = \{c\}$. With $\delta = \delta_0$, any $x \in S$ with $|x-c| < \delta$ equals c, so $|f(x) - f(c)| = 0 < \epsilon$.

2. The two definitions differ only at the point $x = c$, which the limit excludes and the continuity condition includes; there $|f(c) - f(c)| = 0 < \epsilon$ holds automatically.

3. ($\Rightarrow$) Let $x_n \to c$ and $\epsilon > 0$. Take δ from continuity and M with $|x_n - c| < \delta$ for $n \geq M$; then $|f(x_n) - f(c)| < \epsilon$ for $n \geq M$.

   ($\Leftarrow$) If f is not continuous at c, there is $\epsilon_0 > 0$ such that for each n some $x_n \in S$ has $|x_n - c| < \frac{1}{n}$ and $|f(x_n) - f(c)| \geq \epsilon_0$. By the squeeze theorem $x_n \to c$, so the hypothesis gives the contradiction

$$0 = \lim_{n\to\infty} |f(x_n) - f(c)| \geq \epsilon_0.$$

□

Unlike the sequential criterion for limits, part 3 allows $x_n = c$.

15.2 Sine, cosine, and polynomials

Theorem 15.2 [Sine and cosine are continuous]. *The functions* $\sin x$ *and* $\cos x$ *are continuous on* $\mathbb{R}$.

Proof. From the unit-circle definitions, $\sin^2 x + \cos^2 x = 1$, and for all $x, a, b \in \mathbb{R}$:

1. $|\sin x| \leq 1$ and $|\cos x| \leq 1$;

2. $|\sin x| \le |x|$ (Figure 7);

3. $\sin(a+b) = \cos a \sin b + \sin a \cos b$ and $\sin a - \sin b = 2\sin\left(\frac{a-b}{2}\right)\cos\left(\frac{a+b}{2}\right)$.

Sine. Let $\epsilon > 0$ and $\delta = \epsilon$. If $|x-c| < \delta$, then

$$|\sin x - \sin c| = 2\left|\sin\frac{x-c}{2}\right|\left|\cos\frac{x+c}{2}\right| \le 2\left|\sin\frac{x-c}{2}\right| \le 2\,\frac{|x-c|}{2} = |x-c| < \epsilon.$$

Cosine. Since $\cos x = \sin(x+\pi/2)$, if $x_n \to c$ then $x_n+\pi/2 \to c+\pi/2$, and continuity of sine gives $\cos x_n = \sin(x_n+\pi/2) \to \sin(c+\pi/2) = \cos c$, so cosine is continuous by Theorem 15.1(3). □

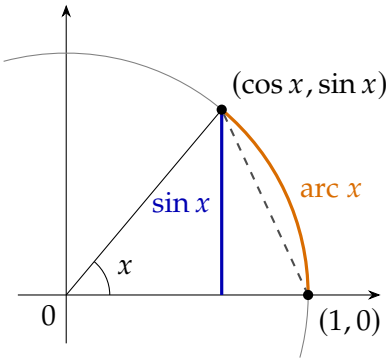


Figure 7: For $0 < x < \pi/2$, the vertical segment of length $\sin x$ is shorter than the chord from $(1,0)$ to $(\cos x, \sin x)$, which is shorter than the arc of length x. Hence $|\sin x| \le |x|$; for $|x| \ge \pi/2$ it follows from $|\sin x| \le 1 < \pi/2$, and oddness of sine handles $x < 0$.

Theorem 15.3 [Polynomials are continuous]. *Every polynomial $f(x) = a_d x^d + \cdots + a_1 x + a_0$ is continuous on $\mathbb{R}$.*

Proof. If $x_n \to c$, the limit laws for sequences give

$$\lim_{n\to\infty} f(x_n) = a_d\left(\lim_n x_n\right)^d + \cdots + a_1 \lim_n x_n + a_0 = a_d c^d + \cdots + a_1 c + a_0 = f(c).$$ □

15.3 Combining continuous functions

Theorem 15.4 [Algebra of continuous functions]. *If $f, g : S \to \mathbb{R}$ are continuous at $c \in S$, then $f+g$ and fg are continuous at c, and if $g(x) \neq 0$ for all $x \in S$, so is f/g.*

Each part follows from Theorem 15.1(3) and the corresponding limit law for sequences, as in the proof for polynomials.

Theorem 15.5 [Composition]. *Let $A, B \subset \mathbb{R}$, $g : A \to B$, and $f : B \to \mathbb{R}$. If g is continuous at c and f is continuous at $g(c)$, then $f \circ g$ is continuous at c.*

Proof. If $x_n \to c$, then $g(x_n) \to g(c)$, and hence $f(g(x_n)) \to f(g(c))$. □

Example: These results give continuity without an ϵ-δ argument.

1. $1/x^2$ is continuous on $(0, \infty)$: it is $1/g(x)$ with $g(x) = x^2$ continuous and nonzero there.
2. $\left(\cos(1/x^2)\right)^2$ is continuous on $(0, \infty)$: cos is continuous on $\mathbb{R}$, so $g(x) = \cos(1/x^2)$ is continuous on $(0, \infty)$, and composing with the continuous $f(x) = x^2$ gives $(f \circ g)(x)$.

15.4 Dirichlet's function

Must every function $f : \mathbb{R} \to \mathbb{R}$ be continuous at some point? No.

Theorem 15.6 [Dirichlet's function]. *The* Dirichlet function

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}, \end{cases}$$

is discontinuous at every $c \in \mathbb{R}$.

Proof. Use Theorem 15.1(3), with the density of the rationals and of the irrationals in $\mathbb{R}$.

1. If $c \in \mathbb{Q}$, choose irrational x_n with $c < x_n < c + \frac{1}{n}$. Then $x_n \to c$, but $f(x_n) = 0$ for all n, so $\lim_n f(x_n) = 0 \neq 1 = f(c)$.
2. If $c \notin \mathbb{Q}$, choose rational x_n with $c < x_n < c + \frac{1}{n}$. Then $x_n \to c$, but $f(x_n) = 1$ for all n, so $\lim_n f(x_n) = 1 \neq 0 = f(c)$. □

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 15, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 16: The Min/Max Theorem and Bolzano's Intermediate Value Theorem

Continuous functions on a closed and bounded interval $[a, b]$ are well behaved: they are bounded, attain their extreme values, and take every value in between, so $f([a, b])$ is itself a closed interval.

16.1 Boundedness and the Min-Max theorem

Definition 16.1 [Bounded function]. A function $f : S \to \mathbb{R}$ is bounded if there is $B \geq 0$ such that $|f(x)| \leq B$ for all $x \in S$.

Theorem 16.2. *If $f : [a, b] \to \mathbb{R}$ is continuous, then f is bounded.*

Proof. Suppose not. Then for every $n \in \mathbb{N}$ there is $x_n \in [a, b]$ with $|f(x_n)| \geq n$. By the Bolzano-Weierstrass theorem, some subsequence $\{x_{n_k}\}$ converges to some x, and $a \leq x_{n_k} \leq b$ gives $x \in [a, b]$. Since f is continuous at x,

$$f(x) = \lim_{k\to\infty} f(x_{n_k}), \qquad \text{so} \qquad |f(x)| = \lim_{k\to\infty} |f(x_{n_k})|.$$

A convergent sequence is bounded, so $\{|f(x_{n_k})|\}$ is bounded, and hence so is $\{n_k\}$, because $n_k \leq |f(x_{n_k})|$. But $k \leq n_k$ for every k, so $\{n_k\}$ is unbounded, a contradiction. □

Definition 16.3 [Absolute minimum and maximum]. Let $f : S \to \mathbb{R}$. Then f achieves an absolute minimum at $c \in S$ if $f(x) \geq f(c)$ for all $x \in S$, and an absolute maximum at $d \in S$ if $f(x) \leq f(d)$ for all $x \in S$.

Theorem 16.4 [Min-Max theorem]. *If $f : [a, b] \to \mathbb{R}$ is continuous, then f achieves an absolute maximum and an absolute minimum on $[a, b]$.*

This is also called the extreme value theorem.

Proof. By Theorem 16.2, the set $E = \{f(x) : x \in [a, b]\}$ is bounded above; let $L = \sup E$. Then $f(x) \leq L$ for all $x \in [a, b]$, and there is a sequence $x_n \in [a, b]$ with $f(x_n) \to L$. By Bolzano-Weierstrass, a subsequence x_{n_k} converges to some $d \in [a, b]$, and by continuity

$$f(d) = \lim_{k\to\infty} f(x_{n_k}) = \lim_{n\to\infty} f(x_n) = L.$$

So f achieves an absolute maximum at d. Applying this to $-f$, which is also continuous, gives an absolute minimum of f. □

Both hypotheses are needed.

- *Continuity.* Define $f : [0,1] \to \mathbb{R}$ by $f(x) = \frac{1}{2}$ for $x = 0, 1$ and $f(x) = x$ for $x \in (0,1)$. Its values fill $(0,1)$, so $\sup f = 1$ and $\inf f = 0$, but neither is attained.
- *A closed and bounded domain.* The function $f(x) = \frac{1}{x} - \frac{1}{1-x}$ is continuous on $S = (0,1)$, but it tends to $+\infty$ as $x \to 0^+$ and to $-\infty$ as $x \to 1^-$, so it has neither an absolute maximum nor an absolute minimum.

So far, if $f : [a,b] \to \mathbb{R}$ is continuous with an absolute minimum at c and an absolute maximum at d, then $f([a,b]) \subseteq [f(c), f(d)]$. The intermediate value theorem shows that every value in this interval is attained.

16.2 Bolzano's intermediate value theorem

Theorem 16.5. *Let $f : [a,b] \to \mathbb{R}$ be continuous. If $f(a) < 0$ and $f(b) > 0$, then there is $c \in (a,b)$ with $f(c) = 0$.*

Proof. Use bisection. Let $a_1 = a$ and $b_1 = b$. Given a_n, b_n, let $m_n = (a_n + b_n)/2$ and set

$$(a_{n+1}, b_{n+1}) = \begin{cases} (a_n, m_n) & \text{if } f(m_n) \geq 0, \\ (m_n, b_n) & \text{if } f(m_n) < 0. \end{cases}$$

By induction, for all n:

1. $a \leq a_n \leq a_{n+1} \leq b_{n+1} \leq b_n \leq b$;
2. $b_{n+1} - a_{n+1} = \frac{b_n - a_n}{2}$, hence $b_n - a_n = \frac{b-a}{2^{n-1}}$;
3. $f(a_n) < 0$ and $f(b_n) \geq 0$.

By (1), $\{a_n\}$ is increasing and $\{b_n\}$ is decreasing, and both are bounded, so $a_n \to c$ and $b_n \to d$ for some $c, d \in [a,b]$. By (2),

$$d - c = \lim_{n\to\infty} (b_n - a_n) = \lim_{n\to\infty} \frac{b-a}{2^{n-1}} = 0,$$

so $c = d$. By continuity and (3), $f(c) = \lim f(a_n) \leq 0$ and $f(c) = \lim f(b_n) \geq 0$, so $f(c) = 0$. Finally $c \neq a$ and $c \neq b$, since $f(a) < 0 < f(b)$. □

Theorem 16.6 [Bolzano's intermediate value theorem]. *Let $f : [a,b] \to \mathbb{R}$ be continuous. If $f(a) < f(b)$ and $y \in (f(a), f(b))$, or $f(b) < f(a)$ and $y \in (f(b), f(a))$, then there is $c \in (a,b)$ with $f(c) = y$.*

This is usually abbreviated IVT.

Proof. Suppose $f(a) < y < f(b)$ and let $g(x) = f(x) - y$. Then g is continuous on $[a, b]$ with $g(a) < 0 < g(b)$, so Theorem 16.5 gives $c \in (a, b)$ with $g(c) = 0$, that is, $f(c) = y$. If $f(b) < y < f(a)$, apply the same argument to $g(x) = y - f(x)$. □

Theorem 16.7. *Let $f : [a, b] \to \mathbb{R}$ be continuous, with an absolute minimum at c and an absolute maximum at d. Then*

$$f([a, b]) = [f(c), f(d)].$$

Proof. We already know $f([a, b]) \subseteq [f(c), f(d)]$. The endpoints $f(c)$ and $f(d)$ are attained, and if $f(c) < f(d)$, the IVT applied to f on the closed interval with endpoints c and d attains every value strictly between them. Hence $[f(c), f(d)] \subseteq f([a, b])$. □

The IVT fails without continuity. In Figure 8, the function equal to x on $[0, 1]$ except at $x = \frac{1}{2}$, where it equals 1, has $f(0) = 0$ and $f(1) = 1$ but never takes the value $\frac{1}{2}$.

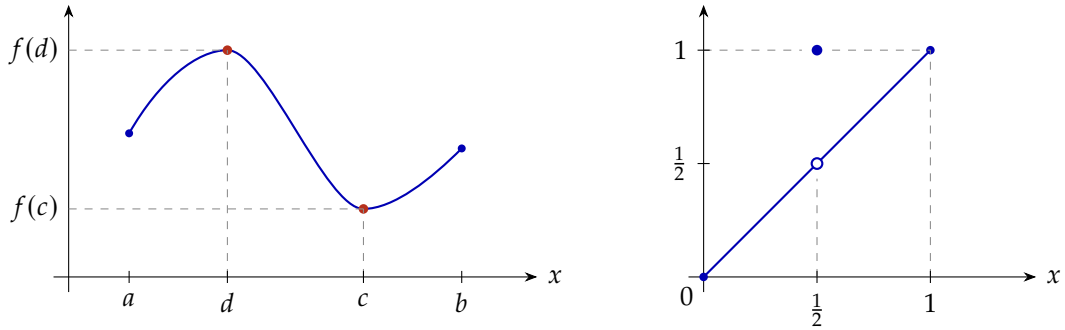


Figure 8: Left: a continuous function on $[a, b]$ with its absolute maximum at d and absolute minimum at c (schematic). Right: $f(x) = x$ for $x \neq \frac{1}{2}$ and $f(\frac{1}{2}) = 1$ on $[0, 1]$; the value $\frac{1}{2}$ is skipped, so the IVT needs continuity.

Example: *A root of a polynomial.* The polynomial $f(x) = x^{2021} + x^{2020} + 9.03x + 1$ has at least one real root. Indeed $f(0) = 1 > 0$ and $f(-1) = -1 + 1 - 9.03 + 1 = -8.03 < 0$, and f is continuous, so by the IVT there is $c \in (-1, 0)$ with $f(c) = 0$.

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 16, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 17: Uniform Continuity and the Definition of the Derivative

In the definition of continuity, δ may depend on both ϵ and the point. Uniform continuity asks for one δ that works everywhere, and on a closed and bounded interval the two notions agree. The second part of the lecture defines the derivative.

17.1 Uniform continuity

Recall that $f : S \to \mathbb{R}$ is continuous on S if for every $c \in S$ and every $\epsilon > 0$ there is $\delta = \delta(\epsilon, c) > 0$ such that for all $x \in S$, $|x - c| < \delta$ implies $|f(x) - f(c)| < \epsilon$.

Example: *$1/x$ is continuous on $(0, 1)$.* Let $c \in (0, 1)$ and $\epsilon > 0$, and choose $\delta = \min\{\frac{c}{2}, \frac{c^2\epsilon}{2}\}$. If $|x - c| < \delta$, then $|x - c| < \frac{c}{2}$, so $|x| > c - |x - c| > \frac{c}{2}$ and $\frac{1}{|x|} < \frac{2}{c}$. Therefore

$$\left|\frac{1}{x} - \frac{1}{c}\right| = \frac{|x - c|}{|x||c|} < \frac{2}{c^2}\,\delta \le \frac{2}{c^2} \cdot \frac{c^2\epsilon}{2} = \epsilon.$$

Here δ depends on both ϵ and c, and it shrinks as $c \to 0$.

Definition 17.1 [Uniform continuity]. A function $f : S \to \mathbb{R}$ is uniformly continuous on S if for every $\epsilon > 0$ there is $\delta = \delta(\epsilon) > 0$ such that for all $x, c \in S$,

$$|x - c| < \delta \implies |f(x) - f(c)| < \epsilon.$$

The only difference is that δ depends on ϵ alone. A uniformly continuous function is continuous.

Example: *x^2 is uniformly continuous on $[0, 1]$.* Given $\epsilon > 0$, take $\delta = \frac{\epsilon}{2}$. For $x, c \in [0, 1]$ with $|x - c| < \delta$,

$$|x^2 - c^2| = |x + c||x - c| \le 2|x - c| < 2\delta = \epsilon.$$

Negating the definition: $f : S \to \mathbb{R}$ is *not* uniformly continuous on S if there is $\epsilon_0 > 0$ such that for every $\delta > 0$ there are $x, c \in S$ with

$$|x - c| < \delta \qquad \text{and} \qquad |f(x) - f(c)| \ge \epsilon_0.$$

Example: *$1/x$ is not uniformly continuous on $(0, 1)$.* Take $\epsilon_0 = 2$ (any $\epsilon_0 > 0$ works). Given $\delta > 0$, let $c = \min\{\delta, \frac{1}{2}\}$ and $x = \frac{c}{2}$. Then $|x - c| = \frac{c}{2} \le \frac{\delta}{2} < \delta$, but

$$\left|\frac{1}{x} - \frac{1}{c}\right| = \frac{2}{c} - \frac{1}{c} = \frac{1}{c} \ge \frac{1}{1/2} = 2.$$

Theorem 17.2. *A function $f : [a, b] \to \mathbb{R}$ is continuous if and only if it is uniformly continuous.*

Proof. Uniform continuity implies continuity: the same $\delta(\epsilon)$ works at every point c. Conversely, suppose f is continuous but not uniformly continuous. Then there is $\epsilon_0 > 0$ such that, taking $\delta = \frac{1}{n}$, for every $n \in \mathbb{N}$ there are $x_n, c_n \in [a, b]$ with

$$|x_n - c_n| < \frac{1}{n} \qquad \text{and} \qquad |f(x_n) - f(c_n)| \geq \epsilon_0.$$

By Bolzano-Weierstrass, a subsequence $\{x_{n_k}\}$ converges to some $x \in [a, b]$. Applying Bolzano-Weierstrass again to the bounded sequence $\{c_{n_k}\}_k$ gives a further subsequence $c_{n_{k_j}} \to c \in [a, b]$, and still $x_{n_{k_j}} \to x$. Then

$$|x - c| = \lim_{j\to\infty} |x_{n_{k_j}} - c_{n_{k_j}}| \leq \lim_{j\to\infty} \frac{1}{n_{k_j}} = 0,$$

so $x = c$. Both $x_{n_{k_j}}$ and $c_{n_{k_j}}$ converge to c, and f is continuous at c, so

$$0 = |f(c) - f(c)| = \lim_{j\to\infty} |f(x_{n_{k_j}}) - f(c_{n_{k_j}})| \geq \epsilon_0,$$

a contradiction. □

17.2 The derivative

Definition 17.3 [Derivative]. Let I be an interval, $f : I \to \mathbb{R}$, and $c \in I$. Then f is DIFFERENTIABLE at c if the limit

$$f'(c) := \lim_{x\to c} \frac{f(x) - f(c)}{x - c}$$

exists; $f'(c)$ is the DERIVATIVE of f at c. If f is differentiable at every $c \in I$, the resulting function is written f' or $\frac{df}{dx}$.

Example: *Affine functions.* If $f(x) = ax + b$, then for every $c \in \mathbb{R}$,

$$f'(c) = \lim_{x\to c} \frac{ax + b - (ac + b)}{x - c} = \lim_{x\to c} a = a.$$

Proposition 17.4 [Power rule]. *For $n \in \mathbb{N}$ and $\alpha \in \mathbb{R}$, the function $f(x) = \alpha x^n$ satisfies $f'(c) = \alpha n c^{n-1}$ for every $c \in \mathbb{R}$.*

Proof. The sum below telescopes: shifting the index $\ell = j + 1$ in the second sum,

$$(x - c) \sum_{j=0}^{n-1} x^{n-1-j} c^j = \sum_{j=0}^{n-1} x^{n-j} c^j - \sum_{\ell=1}^{n} x^{n-\ell} c^\ell = x^n - c^n.$$

Hence, for $x \neq c$, the difference quotient is a polynomial in x, and polynomials are continuous, so

$$\lim_{x \to c} \frac{\alpha x^n - \alpha c^n}{x - c} = \alpha \lim_{x \to c} \sum_{j=0}^{n-1} x^{n-1-j} c^j = \alpha \sum_{j=0}^{n-1} c^{n-1-j} c^j = \alpha n c^{n-1}. \qquad \square$$

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Lecture 18: Weierstrass's Example of a Continuous and Nowhere Differentiable Function

Differentiability implies continuity, but not conversely, and the failure can be total: Weierstrass built a continuous function on $\mathbb{R}$ that is differentiable nowhere, as a sum of increasingly fast oscillations.

18.1 Differentiability and continuity

Theorem 18.1. *If $f : I \to \mathbb{R}$ is differentiable at $c \in I$, then f is continuous at c.*

Proof. Every point of an interval is a cluster point of it, so continuity at c means $\lim_{x\to c} f(x) = f(c)$. For $x \neq c$,

$$\lim_{x\to c} f(x) = \lim_{x\to c}\left((x-c)\,\frac{f(x)-f(c)}{x-c} + f(c)\right) = 0 \cdot f'(c) + f(c) = f(c). \qquad \square$$

Example: *$|x|$ is not differentiable at 0.* It suffices to find $x_n \to 0$, $x_n \neq 0$, along which the difference quotients have no limit. Take $x_n = \frac{(-1)^n}{n}$. Then

$$\frac{f(x_n)-f(0)}{x_n - 0} = \frac{|(-1)^n/n|}{(-1)^n/n} = (-1)^n,$$

which does not converge.

18.2 Three preliminary results

Lemma 18.2 [Cosine estimates].

1. *For all $x, y \in \mathbb{R}$, $|\cos x - \cos y| \leq |x - y|$.*
2. *For every $c \in \mathbb{R}$ and $K \in \mathbb{N}$ there is $y \in (c + \frac{\pi}{K}, c + \frac{3\pi}{K})$ with $|\cos(Kc) - \cos(Ky)| \geq 1$.*

Proof. (1) The proof of continuity of sin (Lecture 15) gave $|\sin x - \sin y| \leq |x - y|$, so $|\cos x - \cos y| = |\sin(x + \frac{\pi}{2}) - \sin(y + \frac{\pi}{2})| \leq |x - y|$.
(2) As y runs over the interval, $t = Ky$ runs over the open interval $(Kc + \pi,\ Kc + 3\pi)$ of length 2π, one full period of cos. On it, $\cos t$ takes every value in $[-1, 1]$, except that the value $\cos(Kc + \pi) = -\cos(Kc)$ at the excluded endpoints is missed when it equals ± 1. Choose y as follows.

- If $0 \le \cos(Kc) < 1$, take $\cos(Ky) = -1$; then $|\cos(Kc) - \cos(Ky)| = 1 + \cos(Kc) \ge 1$.
- If $-1 < \cos(Kc) < 0$, take $\cos(Ky) = 1$; the difference is $1 - \cos(Kc) > 1$.
- If $\cos(Kc) = \pm 1$, take $\cos(Ky) = 0$; the difference is 1. □

Lemma 18.3 [A triangle inequality]. *For all $a, b, c \in \mathbb{R}$, $|a + b + c| \ge |a| - |b| - |c|$.*

Proof. By the triangle inequality, $|a| = |(a + b + c) + (-b) + (-c)| \le |a + b + c| + |b| + |c|$. □

Proposition 18.4 [Weierstrass's function].

1. *For every $x \in \mathbb{R}$, the series $\sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}$ converges absolutely.*
2. *The function $f : \mathbb{R} \to \mathbb{R}$,*
$$f(x) = \sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k},$$
is bounded and continuous.

Proof. (1) Each term satisfies $\left|\frac{\cos(160^k x)}{4^k}\right| \le 4^{-k}$, and $\sum 4^{-k}$ converges, so the comparison test applies.
(2) For all x, $|f(x)| \le \sum_{k=0}^{\infty} 4^{-k} = \frac{4}{3}$.
For continuity, let $c \in \mathbb{R}$ and $x_n \to c$. The sequence $\{|f(x_n) - f(c)|\}$ is bounded, and a nonnegative sequence tends to 0 if and only if its lim sup is 0. So it suffices to show $\limsup_n |f(x_n) - f(c)| \le \epsilon$ for every $\epsilon > 0$. Choose M_0 with $\sum_{k=M_0+1}^{\infty} 4^{-k} < \frac{\epsilon}{2}$. Split the series at M_0, bound the head with Lemma 18.2(1), $|\cos(160^k x_n) - \cos(160^k c)| \le 160^k |x_n - c|$, and bound each tail term by $2 \cdot 4^{-k}$:

$$\limsup_{n\to\infty} |f(x_n) - f(c)| \le \limsup_{n\to\infty} \Big(\sum_{k=0}^{M_0} 40^k \Big) |x_n - c| + 2 \sum_{k=M_0+1}^{\infty} 4^{-k} < 0 + \epsilon.$$ □

18.3 Weierstrass's theorem

Theorem 18.5 [Weierstrass]. *The continuous function $f(x) = \sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}$ is differentiable at no point of $\mathbb{R}$.*

Proof. Fix $c \in \mathbb{R}$. We construct $x_n \to c$, $x_n \ne c$, such that the difference quotients $\frac{f(x_n) - f(c)}{x_n - c}$ are unbounded. If $f'(c)$ existed, these quotients would converge to it and hence be bounded.

By Lemma 18.2(2) with $K = 160^n$, for every n there is x_n with

$$\text{(a)}\ \ \frac{\pi}{160^n} < x_n - c < \frac{3\pi}{160^n}, \qquad \text{(b)}\ \ |\cos(160^n c) - \cos(160^n x_n)| \geq 1.$$

By (a), $x_n \neq c$ and $x_n \to c$. Write $f_k(x) = 4^{-k}\cos(160^k x)$ and split

$$f(c) - f(x_n) = \underbrace{f_n(c) - f_n(x_n)}_{a_n} + \underbrace{\sum_{k=0}^{n-1}\big(f_k(c) - f_k(x_n)\big)}_{b_n} + \underbrace{\sum_{k=n+1}^{\infty}\big(f_k(c) - f_k(x_n)\big)}_{c_n}.$$

By Lemma 18.3, $|f(c) - f(x_n)| \geq |a_n| - |b_n| - |c_n|$. The single term a_n is large by (b):

$$|a_n| = 4^{-n}|\cos(160^n c) - \cos(160^n x_n)| \geq 4^{-n}.$$

The earlier terms oscillate slowly, so Lemma 18.2(1) and (a) make b_n small:

$$|b_n| \leq \sum_{k=0}^{n-1} 4^{-k}\,160^k |x_n - c| \leq \frac{3\pi}{160^n}\sum_{k=0}^{n-1} 40^k = \frac{3\pi}{160^n}\cdot\frac{40^n - 1}{39} \leq \frac{\pi}{13}\,4^{-n} \leq \frac{4}{13}\,4^{-n}.$$

The later terms are small because of their coefficients:

$$|c_n| \leq 2\sum_{k=n+1}^{\infty} 4^{-k} = 2\cdot 4^{-n-1}\cdot\frac{4}{3} = \frac{2}{3}\,4^{-n}.$$

Therefore

$$|f(c) - f(x_n)| \geq 4^{-n}\left(1 - \frac{4}{13} - \frac{2}{3}\right) = \frac{4^{-n}}{39},$$

and, using $|x_n - c| < 3\pi/160^n$,

$$\left|\frac{f(x_n) - f(c)}{x_n - c}\right| \geq \frac{160^n}{3\pi}\cdot\frac{4^{-n}}{39} = \frac{40^n}{117\pi} \longrightarrow \infty. \qquad \square$$

So there is a continuous function $f : \mathbb{R} \to \mathbb{R}$ that is differentiable at no point.

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Lecture 19: Differentiation Rules, Rolle's Theorem, and the Mean Value Theorem

The usual differentiation rules follow from limit laws. The mean value theorem, proved through Rolle's theorem, turns information about f' into information about f.

19.1 Differentiation rules

Theorem 19.1. *Let $f, g : I \to \mathbb{R}$ be differentiable at $c \in I$.*

1. *(Linearity) For all $\alpha \in \mathbb{R}$, $(\alpha f + g)'(c) = \alpha f'(c) + g'(c)$.*
2. *(Product rule) $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$.*
3. *(Quotient rule) If $g(x) \neq 0$ for all $x \in I$, then*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$$

Proof. (1) The difference quotient of $\alpha f + g$ is $\alpha \frac{f(x)-f(c)}{x-c} + \frac{g(x)-g(c)}{x-c}$, which tends to $\alpha f'(c) + g'(c)$.
(2) Write

$$\frac{f(x)g(x) - f(c)g(c)}{x-c} = \frac{f(x)-f(c)}{x-c}\, g(x) + f(c)\, \frac{g(x)-g(c)}{x-c}.$$

Since g is differentiable at c, it is continuous there (Lecture 18), so $g(x) \to g(c)$ and the right side tends to $f'(c)g(c) + f(c)g'(c)$.
(3) Left as an exercise. □

Theorem 19.2 [Chain rule]. *Let I_1, I_2 be intervals, let $g : I_1 \to I_2$ be differentiable at $c \in I_1$, and let $f : I_2 \to \mathbb{R}$ be differentiable at $g(c)$. Then $f \circ g : I_1 \to \mathbb{R}$ is differentiable at c and*

$$(f \circ g)'(c) = f'(g(c))\, g'(c).$$

Proof. Let $h = f \circ g$ and $d = g(c)$. Dividing by $g(x) - g(c)$ directly fails when $g(x) = g(c)$ for x near c, so define

$$u(y) = \begin{cases} \dfrac{f(y)-f(d)}{y-d} & y \neq d, \\ f'(d) & y = d, \end{cases} \qquad v(x) = \begin{cases} \dfrac{g(x)-g(c)}{x-c} & x \neq c, \\ g'(c) & x = c. \end{cases}$$

By the definition of the derivative, $\lim_{y\to d} u(y) = f'(d) = u(d)$ and $\lim_{x\to c} v(x) = g'(c) = v(c)$, so u is continuous at d and v is continuous at c. For all y and x,

$$f(y) - f(d) = u(y)(y-d), \qquad g(x) - g(c) = v(x)(x-c),$$

and therefore

$$h(x) - h(c) = f(g(x)) - f(d) = u(g(x))\big(g(x) - g(c)\big) = u(g(x))\, v(x)\,(x-c).$$

Since g is continuous at c and u is continuous at $g(c)$, the composition $u \circ g$ is continuous at c. Hence

$$\lim_{x\to c} \frac{h(x)-h(c)}{x-c} = \lim_{x\to c} u(g(x))\, v(x) = u(g(c))\, v(c) = f'(g(c))\, g'(c). \qquad \square$$

19.2 Relative extrema

Definition 19.3 [Relative maximum and minimum]. Let $S \subseteq \mathbb{R}$ and $f : S \to \mathbb{R}$. Then f has a relative maximum at $c \in S$ if there is $\delta > 0$ such that for all $x \in S$, $|x - c| < \delta$ implies $f(x) \le f(c)$. A relative minimum is defined with $f(x) \ge f(c)$.

An absolute maximum is a relative maximum, but a relative maximum need not be absolute.

Theorem 19.4. *If $f : [a,b] \to \mathbb{R}$ has a relative maximum or minimum at $c \in (a,b)$ and f is differentiable at c, then $f'(c) = 0$.*

Proof. Suppose f has a relative maximum at c. Shrinking δ, we may assume $(c-\delta, c+\delta) \subseteq (a,b)$ and $f(x) \le f(c)$ on this interval. The points $x_n = c - \frac{\delta}{2n} \in (c-\delta, c)$ tend to c from the left, where the difference quotients are ≥ 0 (numerator ≤ 0, denominator < 0). The points $y_n = c + \frac{\delta}{2n}$ tend to c from the right, where the quotients are ≤ 0. Hence

$$f'(c) = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} \ge 0 \qquad \text{and} \qquad f'(c) = \lim_{n\to\infty} \frac{f(y_n) - f(c)}{y_n - c} \le 0,$$

so $f'(c) = 0$. For a relative minimum, apply this to $-f$. $\square$

19.3 Rolle's theorem and the mean value theorem

Theorem 19.5 [Rolle]. *Let $f : [a,b] \to \mathbb{R}$ be continuous and differentiable on (a,b). If $f(a) = f(b)$, then there is $c \in (a,b)$ with $f'(c) = 0$.*

Proof. Let $K = f(a) = f(b)$. By the Min-Max theorem (Lecture 16), f achieves an absolute maximum at some $c_1 \in [a,b]$ and an absolute minimum at some $c_2 \in [a,b]$.

- If $f(c_1) > K$, then $c_1 \neq a, b$, so $c_1 \in (a,b)$ and $f'(c_1) = 0$ by Theorem 19.4.
- If $f(c_2) < K$, then likewise $c_2 \in (a,b)$ and $f'(c_2) = 0$.
- Otherwise $f(c_1) \leq K \leq f(c_2)$, so the maximum is at most the minimum, $f \equiv K$, and $f'(c) = 0$ for every $c \in (a,b)$. □

Theorem 19.6 [Mean value theorem]. *Let $f : [a,b] \to \mathbb{R}$ be continuous and differentiable on (a,b). Then there is $c \in (a,b)$ with*

$$f(b) - f(a) = f'(c)(b-a).$$

This is often abbreviated MVT. Geometrically, some tangent line is parallel to the secant through $(a, f(a))$ and $(b, f(b))$; see Figure 9.

Proof. Subtract the secant line: let

$$g(x) = f(x) - f(b) + \frac{f(b)-f(a)}{b-a}(b-x).$$

Then g is continuous on $[a,b]$, differentiable on (a,b), and $g(a) = g(b) = 0$. By Rolle's theorem there is $c \in (a,b)$ with

$$0 = g'(c) = f'(c) - \frac{f(b)-f(a)}{b-a}.$$ □

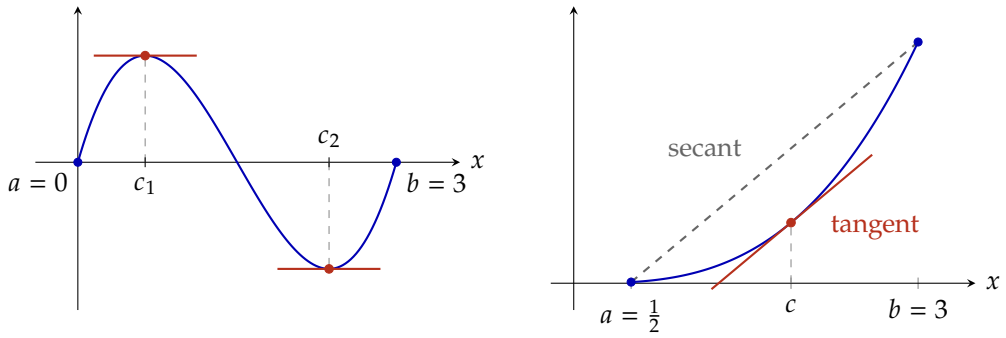


Figure 9: Left: Rolle's theorem for $f(x) = x(x - \frac{3}{2})(x-3)$ on $[0,3]$, with horizontal tangents at $c_{1,2} = \frac{3 \mp \sqrt{3}}{2}$. Right: the mean value theorem for $f(x) = \frac{x^3}{5}$ on $[\frac{1}{2}, 3]$; the tangent at $c = \frac{\sqrt{129}}{6} \approx 1.89$ is parallel to the secant.

19.4 Consequences of the mean value theorem

Theorem 19.7. *If $f : I \to \mathbb{R}$ is differentiable and $f'(x) = 0$ for all $x \in I$, then f is constant.*

Proof. Let $a, b \in I$ with $a < b$. Then f is continuous on $[a, b]$ and differentiable on (a, b), so by the MVT there is $c \in (a, b)$ with $f(b) - f(a) = (b - a)f'(c) = 0$. □

Theorem 19.8. *Let $f : I \to \mathbb{R}$ be differentiable. Then*

1. *f is increasing if and only if $f'(x) \geq 0$ for all $x \in I$;*
2. *f is decreasing if and only if $f'(x) \leq 0$ for all $x \in I$.*

Proof. (1) Suppose $f' \geq 0$ on I, and let $a < b$ in I. By the MVT, $f(b) - f(a) = (b - a)f'(c) \geq 0$ for some $c \in (a, b)$, so $f(a) \leq f(b)$.
Conversely, suppose f is increasing and $c \in I$. For $x \in I$ with $x \neq c$, the differences $f(x) - f(c)$ and $x - c$ have the same sign (or the first is 0), so every difference quotient is ≥ 0. Taking $x_n \to c$ with $x_n \neq c$ from either side gives $f'(c) \geq 0$.
(2) f is decreasing if and only if $-f$ is increasing; apply (1) to $-f$. □

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Lecture 20: Taylor's Theorem and the Definition of Riemann Sums

Taylor's theorem is the mean value theorem for higher derivatives. The second half of the lecture sets up the Riemann integral: partitions, tags, and Riemann sums.

20.1 Taylor's theorem

Definition 20.1 [n times differentiable]. A function $f : I \to \mathbb{R}$ is n times differentiable on $J \subseteq I$ if $f', f'', \ldots, f^{(n)}$ exist at every point of J. Here $f^{(n)}$ denotes the nth derivative of f.

Theorem 20.2 [Taylor]. *Suppose $f : [a, b] \to \mathbb{R}$ has n continuous derivatives on $[a, b]$ and $f^{(n+1)}$ exists on (a, b). Given $x_0, x \in [a, b]$, there is c strictly between x_0 and x (any c if $x = x_0$) such that*

$$f(x) = \underbrace{\sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k}_{P_n(x)} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}}_{R_n(x)}.$$

P_n is the nth order Taylor polynomial of f at x_0, and R_n is the nth order remainder term. For $n = 0$ this is the mean value theorem.

Proof. If $x = x_0$, there is nothing to prove, so let $x \neq x_0$. Define the constant M_{x,x_0} by

$$f(x) = P_n(x) + M_{x,x_0}(x - x_0)^{n+1}, \qquad \text{that is,} \qquad M_{x,x_0} = \frac{f(x) - P_n(x)}{(x - x_0)^{n+1}}.$$

It suffices to show $M_{x,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!}$ for some c between x_0 and x. Differentiating P_n term by term gives $P_n^{(k)}(x_0) = f^{(k)}(x_0)$ for $0 \leq k \leq n$. Let

$$g(s) = f(s) - P_n(s) - M_{x,x_0}(s - x_0)^{n+1}.$$

Then $g^{(k)}$ is continuous on $[a, b]$ for $k \leq n$, $g^{(n+1)}$ exists on (a, b), and

$$g^{(k)}(x_0) = f^{(k)}(x_0) - P_n^{(k)}(x_0) - M_{x,x_0}\frac{(n+1)!}{(n+1-k)!}(x_0 - x_0)^{n+1-k} = 0, \qquad 0 \leq k \leq n.$$

Apply Rolle's theorem repeatedly. Since $g(x_0) = 0$ and, by the choice of M_{x,x_0}, $g(x) = 0$, there is x_1 strictly between x_0 and x with $g'(x_1) = 0$. Since $g'(x_0) = g'(x_1) = 0$, there is x_2 strictly between x_0 and x_1 with $g''(x_2) = 0$. Continuing, we find x_n between x_0 and x_{n-1} with $g^{(n)}(x_n) = 0$, and finally, since $g^{(n)}(x_0) = g^{(n)}(x_n) = 0$, a point c between x_0 and x_n, hence strictly

between x_0 and x, with $g^{(n+1)}(c) = 0$.
Now $P_n^{(n+1)} = 0$ because P_n has degree at most n, and $\frac{d^{n+1}}{ds^{n+1}} M_{x,x_0}(s - x_0)^{n+1} = M_{x,x_0}(n+1)!$. So

$$0 = g^{(n+1)}(c) = f^{(n+1)}(c) - M_{x,x_0}(n+1)!,$$

which gives $M_{x,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!}$. □

Theorem 20.3 [Second derivative test]. *Suppose $f : (a,b) \to \mathbb{R}$ has two continuous derivatives. If $x_0 \in (a,b)$, $f'(x_0) = 0$, and $f''(x_0) > 0$, then f has a strict relative minimum at x_0: there is $\delta > 0$ with $f(x) > f(x_0)$ for $0 < |x - x_0| < \delta$.*

Proof. Since f'' is continuous at x_0 and $f''(x_0) > 0$, there is $\delta > 0$ with $[x_0 - \delta, x_0 + \delta] \subseteq (a,b)$ and $f''(c) > 0$ for all $c \in (x_0 - \delta, x_0 + \delta)$. Let $0 < |x - x_0| < \delta$. Taylor's theorem with $n = 1$ on $[x_0 - \delta, x_0 + \delta]$ gives c between x and x_0, hence in $(x_0 - \delta, x_0 + \delta)$, with

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2 = f(x_0) + \frac{f''(c)}{2}(x - x_0)^2 > f(x_0). \qquad \square$$

20.2 Partitions, tags, and Riemann sums

Riemann integration is the first rigorous theory of area that agrees with the familiar areas of rectangles, triangles, and circles, and it inverts differentiation. It is not a complete theory of area; Lebesgue integration goes further. Write

$$C([a,b]) := \{f : [a,b] \to \mathbb{R} \mid f \text{ is continuous}\}.$$

Definition 20.4 [Partition]. A partition $\underline{x}$ of $[a,b]$ is a finite set

$$\underline{x} = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Its norm is the length of its longest subinterval,

$$\|\underline{x}\| := \max\{x_1 - x_0,\ x_2 - x_1,\ \ldots,\ x_n - x_{n-1}\}.$$

Definition 20.5 [Tag]. A tag of a partition $\underline{x}$ is a finite set $\underline{\xi} = \{\xi_1, \ldots, \xi_n\}$ with

$$a = x_0 \le \xi_1 \le x_1 \le \xi_2 \le x_2 \le \cdots \le x_{n-1} \le \xi_n \le x_n = b,$$

one point in each subinterval. The pair $(\underline{x}, \underline{\xi})$ is a tagged partition.

Example: *A tagged partition of* $[1,3]$. Take $\underline{x} = \{1, \frac{3}{2}, 2, 3\}$ and $\underline{\xi} = \{\frac{5}{4}, \frac{7}{4}, \frac{5}{2}\}$. Then

$$\|\underline{x}\| = \max\left\{\tfrac{3}{2} - 1,\ 2 - \tfrac{3}{2},\ 3 - 2\right\} = 1.$$

Definition 20.6 [Riemann sum]. The Riemann sum of $f : [a,b] \to \mathbb{R}$ for the tagged partition $(\underline{x}, \underline{\xi})$ is

$$S_f(\underline{x}, \underline{\xi}) := \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}).$$

For positive $f \in C([a,b])$, $S_f(\underline{x}, \underline{\xi})$ is the total area of rectangles with base $[x_{k-1}, x_k]$ and height $f(\xi_k)$, an approximation of the area under the graph (Figure 10). As $\|\underline{x}\| \to 0$, these approximations should converge to a number A, interpreted as the area under f on $[a,b]$. Whether they actually converge is the subject of the next lectures.

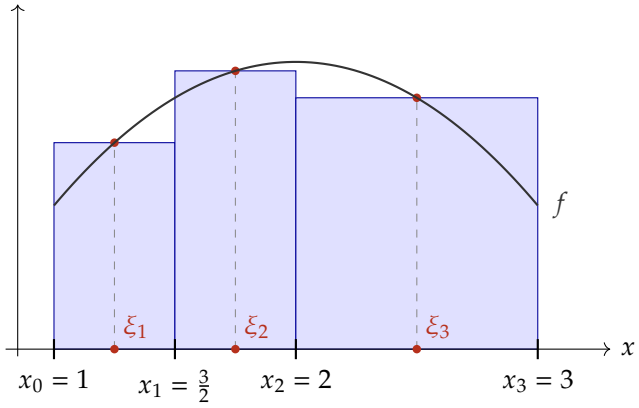


Figure 10: The tagged partition $\underline{x} = \{1, \frac{3}{2}, 2, 3\}$, $\underline{\xi} = \{\frac{5}{4}, \frac{7}{4}, \frac{5}{2}\}$ of $[1,3]$ and the Riemann sum rectangles of heights $f(\xi_k)$, drawn for a schematic positive continuous f.

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Lecture 21: The Riemann Integral of a Continuous Function

Recall the setting of Lecture 20. A partition of $[a,b]$ is a finite set $\underline{x} = \{a = x_0 < x_1 < \cdots < x_n = b\}$ with norm $\|\underline{x}\| = \max_k(x_k - x_{k-1})$. A tag of $\underline{x}$ is $\underline{\xi} = \{\xi_1, \ldots, \xi_n\}$ with $x_{k-1} \le \xi_k \le x_k$, and the pair $(\underline{x}, \underline{\xi})$ is a tagged partition. The Riemann sum of f is

$$S_f(\underline{x}, \underline{\xi}) = \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}),$$

and $C([a,b])$ is the set of continuous $f : [a,b] \to \mathbb{R}$.

Theorem 21.1 [Riemann integral]. *Let $f \in C([a,b])$. There is a unique number $\int_a^b f(x)\,dx \in \mathbb{R}$ such that for every sequence of tagged partitions $\{(\underline{x}^r, \underline{\xi}^r)\}_r$ with $\|\underline{x}^r\| \to 0$,*

$$\lim_{r\to\infty} S_f(\underline{x}^r, \underline{\xi}^r) = \int_a^b f(x)\,dx.$$

Uniqueness is immediate from uniqueness of limits of real sequences, so only existence needs proof. It rests on three estimates, labeled I, II, and III.

21.1 The modulus of continuity

Definition 21.2 [Modulus of continuity]. For $f \in C([a,b])$ and $\eta > 0$, the modulus of continuity of f is

$$w_f(\eta) = \sup\{|f(x) - f(y)| : x, y \in [a,b],\ |x - y| \le \eta\}.$$

The supremum is finite because a continuous function on $[a,b]$ is bounded. Enlarging η enlarges the set, so w_f is nondecreasing.

Theorem 21.3 [Estimate I]. *For every $f \in C([a,b])$, $\lim_{\eta\to 0} w_f(\eta) = 0$: for every $\epsilon > 0$ there is $\delta > 0$ with $w_f(\eta) < \epsilon$ for all $0 < \eta < \delta$.*

Proof. A continuous function on a closed bounded interval is uniformly continuous (Lecture 17). So given $\epsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \epsilon/2$ whenever $|x - y| < \delta$. If $\eta < \delta$ and $|x - y| \le \eta$, then $|x - y| < \delta$, so $\epsilon/2$ is an upper bound for the set defining $w_f(\eta)$. Hence $w_f(\eta) \le \epsilon/2 < \epsilon$. □

21.2 Refinements

Definition 21.4 [Refinement]. If $\underline{x}$ and $\underline{x}'$ are partitions of $[a,b]$ with $\underline{x} \subset \underline{x}'$, then $\underline{x}'$ is a refinement of $\underline{x}$.

A refinement is obtained by adding partition points, so each interval $[x_{k-1}, x_k]$ of $\underline{x}$ is split into consecutive intervals of $\underline{x}'$.

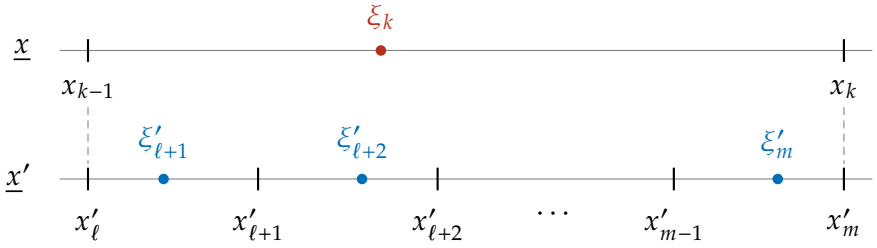


Figure 11: One interval $[x_{k-1}, x_k]$ of $\underline{x}$, with tag ξ_k, and the points of a refinement $\underline{x}'$ in it, with their tags ξ'_j. All the tags lie in $[x_{k-1}, x_k]$, so their values of f differ by at most $w_f(\|\underline{x}\|)$. Schematic.

Theorem 21.5 [Estimate II]. *Let $f \in C([a,b])$, and let $(\underline{x}, \underline{\xi})$ and $(\underline{x}', \underline{\xi}')$ be tagged partitions of $[a,b]$ with $\underline{x}'$ a refinement of $\underline{x}$. Then*

$$|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| \le w_f(\|\underline{x}\|)(b-a).$$

Proof. Fix $k \in \{1, \dots, n\}$ and write the points of $\underline{x}'$ in $[x_{k-1}, x_k]$ as $x_{k-1} = x'_\ell < x'_{\ell+1} < \cdots < x'_m = x_k$, with tags $\xi'_{\ell+1}, \dots, \xi'_m$ (Figure 11). The corresponding piece of $S_f(\underline{x}', \underline{\xi}')$ is $\sum_{j=\ell+1}^{m} f(\xi'_j)(x'_j - x'_{j-1})$. The lengths telescope, $\sum_{j=\ell+1}^{m}(x'_j - x'_{j-1}) = x_k - x_{k-1}$, so

$$\left| f(\xi_k)(x_k - x_{k-1}) - \sum_{j=\ell+1}^{m} f(\xi'_j)(x'_j - x'_{j-1}) \right| = \left| \sum_{j=\ell+1}^{m} \big(f(\xi_k) - f(\xi'_j)\big)(x'_j - x'_{j-1}) \right|.$$

Both ξ_k and ξ'_j lie in $[x_{k-1}, x_k]$, so $|\xi_k - \xi'_j| \le x_k - x_{k-1} \le \|\underline{x}\|$ and $|f(\xi_k) - f(\xi'_j)| \le w_f(\|\underline{x}\|)$, since w_f is nondecreasing. The triangle inequality bounds the last display by $w_f(\|\underline{x}\|)(x_k - x_{k-1})$. The difference of the two Riemann sums is the sum over k of these pieces, so

$$|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| \le \sum_{k=1}^{n} w_f(\|\underline{x}\|)(x_k - x_{k-1}) = w_f(\|\underline{x}\|)(b-a). \qquad \square$$

Theorem 21.6 [Estimate III]. *If $f \in C([a,b])$ and $(\underline{x}, \underline{\xi})$, $(\underline{x}', \underline{\xi}')$ are any two tagged partitions of $[a,b]$, then*

$$|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| \le \big(w_f(\|\underline{x}\|) + w_f(\|\underline{x}'\|)\big)(b-a).$$

Proof. Let $\underline{x}'' = \underline{x} \cup \underline{x}'$, a common refinement, with any tag $\underline{\xi}''$. By the triangle inequality and Theorem 21.5 applied twice,

$$\begin{aligned} |S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| &\le |S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}'', \underline{\xi}'')| + |S_f(\underline{x}', \underline{\xi}') - S_f(\underline{x}'', \underline{\xi}'')| \\ &\le w_f(\|\underline{x}\|)(b-a) + w_f(\|\underline{x}'\|)(b-a). \end{aligned}$$

□

21.3 Existence of the integral

Proof of Theorem 21.1. Fix one sequence of tagged partitions $(\underline{y}(r), \underline{\zeta}(r))$ with $\|\underline{y}(r)\| \to 0$.
Claim 1: $\{S_f(\underline{y}(r), \underline{\zeta}(r))\}_r$ is Cauchy. Let $\epsilon > 0$. By Theorem 21.3 there is $\delta > 0$ with $w_f(\eta) < \epsilon/(2(b-a))$ for all $\eta < \delta$, and there is M with $\|\underline{y}(r)\| < \delta$ for all $r \ge M$. For $r, s \ge M$, Theorem 21.6 gives

$$\begin{aligned} |S_f(\underline{y}(r), \underline{\zeta}(r)) - S_f(\underline{y}(s), \underline{\zeta}(s))| &\le \big(w_f(\|\underline{y}(r)\|) + w_f(\|\underline{y}(s)\|)\big)(b-a) \\ &< \left(\frac{\epsilon}{2(b-a)} + \frac{\epsilon}{2(b-a)}\right)(b-a) = \epsilon. \end{aligned}$$

Cauchy sequences of real numbers converge, so $L = \lim_{r\to\infty} S_f(\underline{y}(r), \underline{\zeta}(r))$ exists.
Claim 2: every sequence $(\underline{x}(r), \underline{\xi}(r))$ with $\|\underline{x}(r)\| \to 0$ has $S_f(\underline{x}(r), \underline{\xi}(r)) \to L$. By the triangle inequality and Theorem 21.6,

$$|S_f(\underline{x}(r), \underline{\xi}(r)) - L| \le \big(w_f(\|\underline{x}(r)\|) + w_f(\|\underline{y}(r)\|)\big)(b-a) + |S_f(\underline{y}(r), \underline{\zeta}(r)) - L|.$$

As $r \to \infty$ the first term tends to 0 by Theorem 21.3 and the second by definition of L. By the squeeze theorem $S_f(\underline{x}(r), \underline{\xi}(r)) \to L$. So L is the required number $\int_a^b f(x)\,dx$. □

21.4 Linearity

We abbreviate $\int_a^b f(x)\,dx$ to $\int_a^b f$.

Theorem 21.7 [Linearity]. *If $f, g \in C([a,b])$ and $\alpha \in \mathbb{R}$, then $\int_a^b (\alpha f + g) = \alpha \int_a^b f + \int_a^b g$.*

Proof. Let $(\underline{x}(r), \underline{\xi}(r))$ be tagged partitions with $\|\underline{x}(r)\| \to 0$. Riemann sums are linear in the function: $S_{\alpha f+g}(\underline{x}(r), \underline{\xi}(r)) = \alpha S_f(\underline{x}(r), \underline{\xi}(r)) + S_g(\underline{x}(r), \underline{\xi}(r))$. Since $\alpha f + g \in C([a,b])$, letting $r \to \infty$ and using the limit laws gives $\int_a^b (\alpha f + g) = \alpha \int_a^b f + \int_a^b g$. □

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Lecture 22: The Fundamental Theorem of Calculus, Integration by Parts, and Change of Variable Formula

Throughout, $f \in C([a,b])$ and $\int_a^b f$ is the Riemann integral of Lecture 21. Each property below is proved by choosing one convenient sequence of tagged partitions $(\underline{x}(r), \underline{\xi}(r))$ with $\|\underline{x}(r)\| \to 0$, comparing Riemann sums, and letting $r \to \infty$.

22.1 Basic properties of the integral

Theorem 22.1 [Additivity]. *If $f \in C([a,b])$ and $a < c < b$, then* $\displaystyle\int_a^b f = \int_a^c f + \int_c^b f.$

Proof. Let $(\underline{y}(r), \underline{\zeta}(r))$ and $(\underline{z}(r), \underline{\eta}(r))$ be tagged partitions of $[a,c]$ and $[c,b]$ with norms tending to 0. Then $\underline{x}(r) = \underline{y}(r) \cup \underline{z}(r)$ with tag $\underline{\xi}(r) = \underline{\zeta}(r) \cup \underline{\eta}(r)$ is a tagged partition of $[a,b]$ with $\|\underline{x}(r)\| = \max\{\|\underline{y}(r)\|, \|\underline{z}(r)\|\} \to 0$, and its Riemann sum splits as $S_f(\underline{x}(r), \underline{\xi}(r)) = S_f(\underline{y}(r), \underline{\zeta}(r)) + S_f(\underline{z}(r), \underline{\eta}(r))$. Let $r \to \infty$. □

Theorem 22.2 [Bounds]. *Let $f \in C([a,b])$, $m_f = \inf_{[a,b]} f$ and $M_f = \sup_{[a,b]} f$. Then*

$$m_f(b-a) \le \int_a^b f \le M_f(b-a).$$

Proof. For any tagged partition, $m_f \sum_k (x_k - x_{k-1}) \le \sum_k f(\xi_k)(x_k - x_{k-1}) \le M_f \sum_k (x_k - x_{k-1})$, and $\sum_k (x_k - x_{k-1}) = b - a$. So $m_f(b-a) \le S_f(\underline{x}(r), \underline{\xi}(r)) \le M_f(b-a)$ for all r; let $r \to \infty$. □

Theorem 22.3 [Monotonicity and triangle inequality]. *Let $f, g \in C([a,b])$.*

1. *If $f(x) \le g(x)$ for all $x \in [a,b]$, then $\int_a^b f \le \int_a^b g$.*
2. $\left|\int_a^b f\right| \le \int_a^b |f|$.

Proof. (1) Termwise, $S_f(\underline{x}(r), \underline{\xi}(r)) \le S_g(\underline{x}(r), \underline{\xi}(r))$ for every r; let $r \to \infty$. (2) The function $|f|$ is continuous and $\pm f \le |f|$, so by (1) and linearity $\pm \int_a^b f \le \int_a^b |f|$, that is, $-\int_a^b |f| \le \int_a^b f \le \int_a^b |f|$. □

We adopt the conventions $\int_a^a f = 0$ and $\int_b^a f = -\int_a^b f$ for $a < b$. The first is consistent with the

theorems so far, since $\left|\int_a^b f\right| \le (b-a)\max|f| \to 0$ as $b \to a$. With them, additivity $\int_a^b f = \int_a^c f + \int_c^b f$ holds for any order of a, b, c in an interval where f is continuous.

22.2 The fundamental theorem of calculus

Theorem 22.4 [Fundamental theorem of calculus]. *Let $f \in C([a,b])$.*

1. *If $F : [a,b] \to \mathbb{R}$ is differentiable and $F' = f$, then $\int_a^b f = F(b) - F(a)$.*
2. *The function $G(x) = \int_a^x f$ is differentiable on $[a,b]$, with $G' = f$ and $G(a) = 0$.*

Proof. (1) Let $\underline{x}(r)$ be partitions with $\|\underline{x}(r)\| \to 0$. By the mean value theorem, for every r and j there is $\xi_j(r) \in [x_{j-1}(r), x_j(r)]$ with

$$F(x_j(r)) - F(x_{j-1}(r)) = F'(\xi_j(r))\big(x_j(r) - x_{j-1}(r)\big) = f(\xi_j(r))\big(x_j(r) - x_{j-1}(r)\big).$$

Use these points as tags. Then the Riemann sum telescopes:

$$\int_a^b f = \lim_{r\to\infty} \sum_j f(\xi_j(r))\big(x_j(r) - x_{j-1}(r)\big) = \lim_{r\to\infty} \sum_j \big(F(x_j(r)) - F(x_{j-1}(r))\big) = F(b) - F(a).$$

(2) $G(a) = 0$ by convention. Fix $c \in [a,b]$; we show $\frac{G(x)-G(c)}{x-c} \to f(c)$ as $x \to c$ (one-sided at an endpoint). By additivity, $G(x) - G(c) = \int_c^x f$, and $\int_c^x f(c)\,dt = f(c)(x-c)$ by Theorem 22.2 applied to a constant. Let $\epsilon > 0$. By continuity of f at c, there is $\delta > 0$ with $|f(t) - f(c)| < \epsilon/2$ whenever $|t - c| < \delta$. If $0 < x - c < \delta$, every $t \in [c,x]$ satisfies $|t-c| < \delta$, so by Theorems 22.3 and 22.2,

$$\left|\frac{G(x)-G(c)}{x-c} - f(c)\right| = \frac{1}{x-c}\left|\int_c^x \big(f(t) - f(c)\big)\,dt\right| \le \frac{1}{x-c}\int_c^x |f(t) - f(c)|\,dt \le \frac{\epsilon}{2} < \epsilon.$$

The case $0 < c - x < \delta$ is the same with the integral over $[x,c]$. □

In particular, every $f \in C([a,b])$ has an antiderivative, namely G.

22.3 Integration by parts and the Riemann–Lebesgue lemma

Theorem 22.5 [Integration by parts]. *If $f, g, f', g' \in C([a,b])$, then*

$$\int_a^b f'g = f(b)g(b) - f(a)g(a) - \int_a^b fg'.$$

Proof. By the product rule $(fg)' = f'g + fg'$, which is continuous. Part 1 of the fundamental theorem and linearity give $f(b)g(b) - f(a)g(a) = \int_a^b f'g + \int_a^b fg'$. □

Definition 22.6 [Fourier coefficients]. For $f \in C([-\pi, \pi])$ and $n \geq 0$, the Fourier coefficients of f are

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx, \qquad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx.$$

Lemma 22.7 [Riemann–Lebesgue]. *Suppose $f, f' \in C([-\pi, \pi])$ and $f(-\pi) = f(\pi)$. Then $a_n \to 0$ and $b_n \to 0$ as $n \to \infty$.*

Proof. For $n \geq 1$, integrate by parts using $\cos(nx) = \left(\frac{1}{n}\sin(nx)\right)'$ and $\sin(nx) = \left(-\frac{1}{n}\cos(nx)\right)'$:

$$b_n = -\frac{1}{n\pi} \int_{-\pi}^{\pi} \sin(nx) f'(x)\, dx, \qquad a_n = \frac{1}{n\pi} \int_{-\pi}^{\pi} \cos(nx) f'(x)\, dx.$$

The boundary terms vanish: for b_n because $\sin(\pm n\pi) = 0$, and for a_n because it equals $-\frac{\cos(n\pi)}{n\pi}\big(f(\pi) - f(-\pi)\big) = 0$. Since $|\sin|, |\cos| \leq 1$, the triangle inequality for integrals gives $|a_n|, |b_n| \leq \frac{1}{n\pi} \int_{-\pi}^{\pi} |f'| \to 0$, and the squeeze theorem finishes the proof. □

22.4 Change of variables

Theorem 22.8 [Change of variables]. *Let $\varphi : [a, b] \to [c, d]$ be continuously differentiable with $\varphi' > 0$ on $[a, b]$, $\varphi(a) = c$, and $\varphi(b) = d$. Then for $f \in C([c, d])$,*

$$\int_c^d f(u)\, du = \int_a^b f(\varphi(x))\varphi'(x)\, dx.$$

Proof. By part 2 of the fundamental theorem, f has an antiderivative $F : [c, d] \to \mathbb{R}$, $F' = f$. By the chain rule, $\big(F(\varphi(x))\big)' = f(\varphi(x))\varphi'(x)$, which is continuous on $[a, b]$. Part 1 of the fundamental theorem, applied on both sides, gives

$$\int_a^b f(\varphi(x))\varphi'(x)\, dx = F(\varphi(b)) - F(\varphi(a)) = F(d) - F(c) = \int_c^d f(u)\, du.$$ □

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 22, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 23: Pointwise and Uniform Convergence of Sequences of Functions

23.1 Power series

Power series motivate the study of sequences of functions.

Definition 23.1 [Power series]. A power series about x_0 is a series of the form $\sum_{m=0}^{\infty} a_m (x - x_0)^m$.

Theorem 23.2 [Radius of convergence]. *Suppose $R = \lim_{m\to\infty} |a_m|^{1/m}$ exists, and let*

$$\rho = \begin{cases} 1/R, & R > 0, \\ \infty, & R = 0. \end{cases}$$

Then $\sum a_m (x - x_0)^m$ converges absolutely if $|x - x_0| < \rho$ and diverges if $|x - x_0| > \rho$.

Proof. For fixed x, $\lim_{m\to\infty} |a_m (x - x_0)^m|^{1/m} = R|x - x_0|$. This is less than 1 when $|x - x_0| < \rho$ and greater than 1 when $|x - x_0| > \rho$, so the root test (Lecture 12) gives both statements. □

Definition 23.3 [Radius of convergence]. The number $\rho \in [0, \infty]$ of Theorem 23.2 is the radius of convergence of the power series.

If $\rho > 0$, the power series defines a function $f : (x_0 - \rho, x_0 + \rho) \to \mathbb{R}$,

$$f(x) = \sum_{m=0}^{\infty} a_m (x - x_0)^m = \lim_{n\to\infty} f_n(x), \qquad f_n(x) = \sum_{m=0}^{n} a_m (x - x_0)^m.$$

So f is a limit of the polynomials f_n. For example, the geometric series gives $\frac{1}{1-x} = \sum_{m=0}^{\infty} x^m$ on $(-1, 1)$. This raises three questions, which drive the rest of the course.

1. Is f continuous?
2. Is f differentiable, and is $f' = \lim_{n\to\infty} f_n'$?
3. Is $\int_a^b f = \lim_{n\to\infty} \int_a^b f_n$?

23.2 Pointwise convergence

We now consider sequences of functions in general, not only partial sums of power series.

Definition 23.4 [Pointwise convergence]. Let $f_n : S \to \mathbb{R}$ for $n \in \mathbb{N}$, and let $f : S \to \mathbb{R}$. The sequence $\{f_n\}$ converges pointwise to f if $\lim_{n\to\infty} f_n(x) = f(x)$ for every $x \in S$.

Example: *Three pointwise limits.*

1. Let $f_n(x) = x^n$ on $[0,1]$. Then

$$\lim_{n\to\infty} f_n(x) = f(x) = \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1. \end{cases}$$

   A pointwise limit of continuous functions need not be continuous.

2. Let $f_n(x) = \sum_{m=0}^{n} x^m$ on $(-1,1)$. Then $f_n(x) \to \frac{1}{1-x}$ pointwise: the partial sums converge to the sum of the power series.

3. Let $f_n : [0,1] \to \mathbb{R}$ be the tent function (Figure 12)

$$f_n(x) = \begin{cases} 4n^2 x, & x \in \left[0, \frac{1}{2n}\right], \\ 4n - 4n^2 x, & x \in \left[\frac{1}{2n}, \frac{1}{n}\right], \\ 0, & x \in \left[\frac{1}{n}, 1\right]. \end{cases}$$

   At $x = 0$, $f_n(0) = 0$ for all n. If $x \in (0,1]$, choose $N \in \mathbb{N}$ with $\frac{1}{N} < x$; then $x > \frac{1}{n}$ for all $n \geq N$, so $f_n(x) = 0$. The sequence $\{f_n(x)\}$ is eventually 0, and $f_n \to 0$ pointwise on $[0,1]$, even though the peaks $f_n\left(\frac{1}{2n}\right) = 2n$ grow without bound.

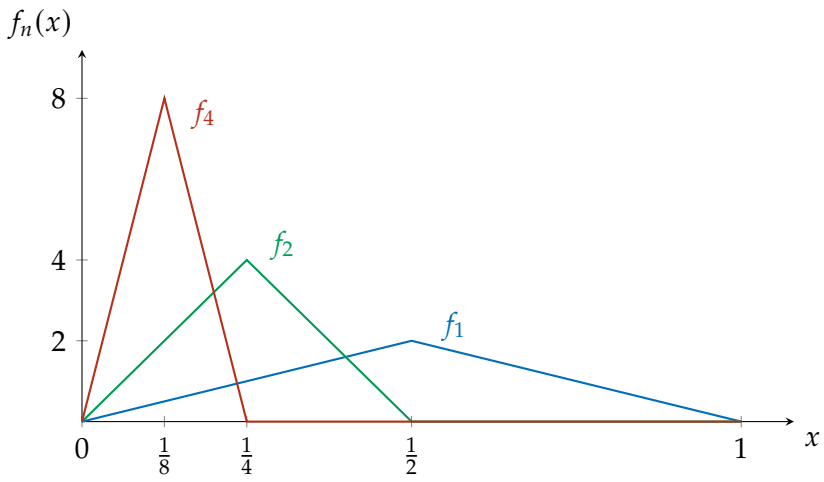


Figure 12: The tent functions f_1, f_2, and f_4. The graph of f_n is a triangle with base $[0, \frac{1}{n}]$ and peak $2n$ at $x = \frac{1}{2n}$; every fixed $x > 0$ eventually lies to the right of the base.

23.3 Uniform convergence

Definition 23.5 [Uniform convergence]. Let $f_n : S \to \mathbb{R}$ for $n \in \mathbb{N}$, and let $f : S \to \mathbb{R}$. The sequence $\{f_n\}$ converges uniformly to f if for every $\epsilon > 0$ there is $M \in \mathbb{N}$ such that for all $n \geq M$ and all $x \in S$,

$$|f_n(x) - f(x)| < \epsilon.$$

In pointwise convergence M may depend on ϵ and x; in uniform convergence one M must work for every $x \in S$ at once.

Theorem 23.6 [Uniform implies pointwise]. *If $f_n \to f$ uniformly on S, then $f_n \to f$ pointwise on S.*

Proof. Let $c \in S$ and $\epsilon > 0$. Uniform convergence gives M with $|f_n(x) - f(x)| < \epsilon$ for all $n \geq M$ and all $x \in S$, in particular $|f_n(c) - f(c)| < \epsilon$ for $n \geq M$. So $f_n(c) \to f(c)$. □

Source. Casey Rodriguez, MIT 18.100A, *Real Analysis*, Lecture 23, Fall 2020. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 24: Uniform Convergence, the Weierstrass M-Test, and Interchanging Limits

24.1 Uniform convergence depends on the domain

To show that $f_n \not\to f$ uniformly on S, negate the definition.

Definition: $f_n : S \to \mathbb{R}$ *does not* converge uniformly to $f : S \to \mathbb{R}$ if there is $\epsilon_0 > 0$ such that for every $M \in \mathbb{N}$ there exist $n \geq M$ and $x \in S$ with $|f_n(x) - f(x)| \geq \epsilon_0$.

Theorem 24.1. *Let $f_n(x) = x^n$ and $f(x) = 0$ for $x \in [0,1)$, $f(1) = 1$.*

1. *For every $0 < b < 1$, $f_n \to f$ uniformly on $[0, b]$.*
2. *f_n does not converge to f uniformly on $[0,1]$.*

Proof. (1) Let $\epsilon > 0$. Since $0 < b < 1$, $b^n \to 0$, so there is M with $b^n < \epsilon$ for all $n \geq M$. Then for $n \geq M$ and every $x \in [0, b]$,

$$|f_n(x) - f(x)| = x^n \leq b^n < \epsilon.$$

(2) Take $\epsilon_0 = \frac{1}{4}$. Given $M \in \mathbb{N}$, let $n = M$ and $x = \left(\frac{1}{2}\right)^{1/M} \in (0,1)$. Then $|f_M(x) - f(x)| = x^M = \frac{1}{2} \geq \epsilon_0$. □

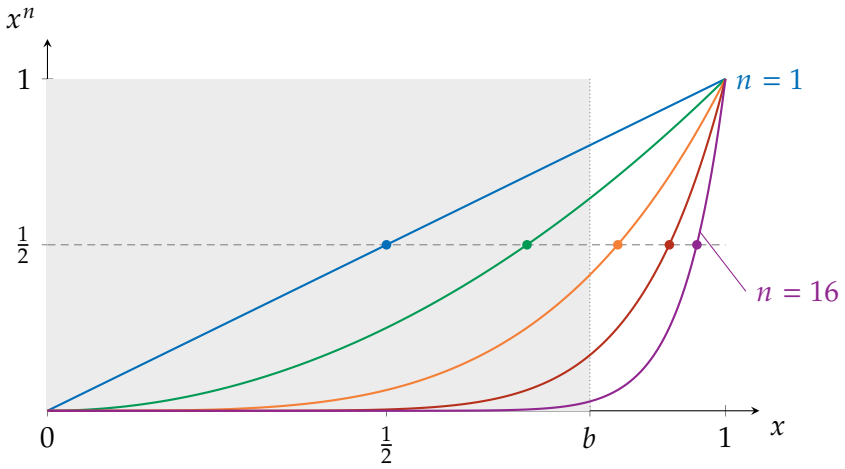


Figure 13: The functions x^n for $n = 1, 2, 4, 8, 16$. On $[0, b]$ (shaded, here $b = 0.8$) every graph lies below b^n, which tends to 0. On $[0,1]$ each x^n still reaches the level $\frac{1}{2}$, at $x = (\frac{1}{2})^{1/n}$ (marked), which moves toward 1.

24.2 The Weierstrass M-test

Theorem 24.2 [Weierstrass M-test]. *Let $f_j : S \to \mathbb{R}$, and suppose there are $M_j > 0$ such that*

(a) $|f_j(x)| \le M_j$ *for all* $x \in S$, *and*

(b) $\sum_{j=1}^{\infty} M_j$ *converges.*

Then

1. $\sum_{j=1}^{\infty} f_j(x)$ *converges absolutely for every* $x \in S$, *and*
2. *with* $f(x) = \sum_{j=1}^{\infty} f_j(x)$, *the partial sums* $\sum_{j=1}^{n} f_j$ *converge to* f *uniformly on* S.

Proof. (1) follows from (a), (b), and the comparison test. (2) Let $\epsilon > 0$. Since $\sum M_j$ converges, its tails tend to 0: there is N with $\sum_{j=n+1}^{\infty} M_j < \epsilon$ for all $n \ge N$. Then for all $n \ge N$ and all $x \in S$,

$$\left| f(x) - \sum_{j=1}^{n} f_j(x) \right| = \left| \sum_{j=n+1}^{\infty} f_j(x) \right| \le \sum_{j=n+1}^{\infty} |f_j(x)| \le \sum_{j=n+1}^{\infty} M_j < \epsilon.$$

The bound does not depend on x, which is exactly uniform convergence. □

24.3 Interchanging limits

In general, limits cannot be interchanged. For example,

$$\lim_{n\to\infty} \lim_{k\to\infty} \frac{n/k}{n/k+1} = \lim_{n\to\infty} \frac{0}{0+1} = 0, \qquad \lim_{k\to\infty} \lim_{n\to\infty} \frac{n/k}{n/k+1} = \lim_{k\to\infty} 1 = 1.$$

The three questions of Lecture 23 are questions of this kind.

1. If each $f_n : S \to \mathbb{R}$ is continuous and $f_n \to f$, is f continuous? That is, is $\lim_{x\to c} \lim_n f_n(x) = \lim_n \lim_{x\to c} f_n(x)$?
2. If each $f_n : [a,b] \to \mathbb{R}$ is differentiable, $f_n \to f$, and $f_n' \to g$, is f differentiable with $f' = g$?
3. If $f_n, f : [a,b] \to \mathbb{R}$ are continuous and $f_n \to f$, is $\int_a^b f_n \to \int_a^b f$?

If the convergence is only pointwise, the answer to all three is no (Theorem 24.1 already shows this for question 1). If the convergence is uniform (for question 2, uniform convergence of the derivatives f_n'), the answer to all three is yes. Lecture 25 proves both claims.

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Lecture 25: Power Series and the Weierstrass Approximation Theorem

25.1 Pointwise convergence does not commute with limits

Lecture 24 asked whether continuity, derivatives, and integrals pass to the limit of a sequence $f_n \to f$. Under pointwise convergence the answer to each question is no.

1. *Continuity.* Each $f_n(x) = x^n$ is continuous on $[0,1]$, but the pointwise limit, 0 on $[0,1)$ and 1 at $x = 1$, is not.
2. *Derivatives.* Let $f_n(x) = \frac{x^{n+1}}{n+1}$ on $[0,1]$. Then $|f_n| \le \frac{1}{n+1}$, so $f_n \to 0$ (even uniformly). But $f_n'(x) = x^n \to g(x)$, where $g = 0$ on $[0,1)$ and $g(1) = 1$. So $g(1) = 1 \neq 0 = (0)'$.
3. *Integrals.* The tent functions f_n of Lecture 23 converge to 0 pointwise on $[0,1]$, but each graph is a triangle with base $\frac{1}{n}$ and height $2n$, so

$$\int_0^1 f_n = \tfrac{1}{2} \cdot \tfrac{1}{n} \cdot 2n = 1 \not\to 0 = \int_0^1 0.$$

25.2 Uniform limits

Under uniform convergence all three answers are yes.

Theorem 25.1 [Continuity of uniform limits]. *If each $f_n : S \to \mathbb{R}$ is continuous and $f_n \to f$ uniformly on S, then f is continuous.*

Proof. Let $c \in S$ and $\epsilon > 0$. Choose M with $|f_n(y) - f(y)| < \epsilon/3$ for all $n \ge M$ and $y \in S$. Since f_M is continuous at c, there is $\delta > 0$ with $|f_M(x) - f_M(c)| < \epsilon/3$ whenever $|x - c| < \delta$. For such x,

$$|f(x) - f(c)| \le |f(x) - f_M(x)| + |f_M(x) - f_M(c)| + |f_M(c) - f(c)| < \tfrac{\epsilon}{3} + \tfrac{\epsilon}{3} + \tfrac{\epsilon}{3} = \epsilon. \qquad \square$$

Theorem 25.2 [Integrals of uniform limits]. *If each $f_n : [a,b] \to \mathbb{R}$ is continuous and $f_n \to f$ uniformly on $[a,b]$, then $\int_a^b f_n \to \int_a^b f$.*

Proof. By Theorem 25.1, f is continuous, so $\int_a^b f$ exists. Let $\epsilon > 0$ and choose M with $|f_n(x) - f(x)| < \frac{\epsilon}{b-a}$ for all $n \ge M$ and $x \in [a,b]$. For $n \ge M$, the triangle inequality and

monotonicity of the integral give

$$\left|\int_a^b f_n - \int_a^b f\right| \le \int_a^b |f_n - f| \le \frac{\epsilon}{b-a}(b-a) = \epsilon. \qquad \square$$

In other words, $\lim_{n\to\infty} \int_a^b f_n = \int_a^b \lim_{n\to\infty} f_n$ for uniformly convergent sequences of continuous functions.

Theorem 25.3 [Derivatives of limits]. *Let each $f_n : [a,b] \to \mathbb{R}$ be continuously differentiable, and let $f, g : [a,b] \to \mathbb{R}$ with $f_n \to f$ pointwise and $f_n' \to g$ uniformly. Then f is continuously differentiable and $f' = g$.*

Proof. By the fundamental theorem of calculus, $f_n(x) - f_n(a) = \int_a^x f_n'$ for all n and $x \in [a,b]$. The limit g is continuous by Theorem 25.1, and Theorem 25.2 on $[a,x]$ gives

$$f(x) - f(a) = \lim_{n\to\infty} \big(f_n(x) - f_n(a)\big) = \lim_{n\to\infty} \int_a^x f_n' = \int_a^x g.$$

So $f(x) = f(a) + \int_a^x g$, and by the fundamental theorem f is differentiable with $f' = g$, which is continuous. $\square$

Example 2 above shows that uniform convergence of the f_n alone is not enough; the derivatives must converge uniformly.

25.3 Power series

Return to a power series $\sum_{j=0}^{\infty} a_j (x - x_0)^j$ with radius of convergence $\rho \in (0, \infty]$, where $\rho^{-1} = \lim_j |a_j|^{1/j}$ as in Lecture 23 (read $\rho^{-1} = 0$ when $\rho = \infty$).

Theorem 25.4 [Uniform convergence of power series]. *For every $r \in (0, \rho)$, the power series converges uniformly on $[x_0 - r, x_0 + r]$.*

Proof. For $x \in [x_0 - r, x_0 + r]$ and $j \ge 0$, $|a_j (x - x_0)^j| \le |a_j| r^j =: M_j$. Then

$$\lim_{j\to\infty} M_j^{1/j} = r \lim_{j\to\infty} |a_j|^{1/j} = \begin{cases} r/\rho, & \rho < \infty, \\ 0, & \rho = \infty, \end{cases}$$

which is less than 1 since $r < \rho$. By the root test $\sum M_j$ converges, and the Weierstrass M-test gives uniform convergence. $\square$

Theorem 25.5 [Termwise differentiation and integration]. *Let $f(x) = \sum_{j=0}^{\infty} a_j (x - x_0)^j$ have radius of convergence $\rho \in (0, \infty]$.*

1. *For every $c \in (x_0 - \rho, x_0 + \rho)$, f is differentiable at c and $f'(c) = \sum_{j=1}^{\infty} j a_j (c - x_0)^{j-1}$.*

2. *For $x_0 - \rho < a < b < x_0 + \rho$,*

$$\int_a^b f = \sum_{j=0}^{\infty} a_j \Big(\frac{(b - x_0)^{j+1}}{j+1} - \frac{(a - x_0)^{j+1}}{j+1} \Big).$$

Proof. The differentiated series is $\sum_{k=0}^{\infty} (k+1) a_{k+1} (x - x_0)^k$, and

$$\big((k+1)|a_{k+1}|\big)^{1/k} = (k+1)^{1/k} \Big(|a_{k+1}|^{1/(k+1)} \Big)^{(k+1)/k} \to 1 \cdot \rho^{-1} = \rho^{-1},$$

so it has the same radius of convergence ρ.
(1) Choose r with $|c - x_0| < r < \rho$ and work on $I = [x_0 - r, x_0 + r]$. The partial sums f_n are polynomials, $f_n \to f$ pointwise on I, and the derivatives f_n' are the partial sums of the differentiated series, which converge uniformly on I by Theorem 25.4. By Theorem 25.3, f is differentiable on I with f' equal to the differentiated series; since c is interior to I, this is the derivative at c.
(2) With $r = \max\{|a - x_0|, |b - x_0|\} < \rho$, $[a, b] \subset [x_0 - r, x_0 + r]$, so $f_n \to f$ uniformly on $[a, b]$. By Theorem 25.2, $\int_a^b f = \lim_n \int_a^b f_n$, and $\int_a^b (x - x_0)^j \, dx = \frac{(b-x_0)^{j+1} - (a-x_0)^{j+1}}{j+1}$ by the fundamental theorem. □

Since the differentiated series again has radius ρ, part 1 applies repeatedly: f is infinitely differentiable on $(x_0 - \rho, x_0 + \rho)$ and $f^{(k)}(x) = \sum_{j \geq k} j(j-1) \cdots (j-k+1) a_j (x - x_0)^{j-k}$. At $x = x_0$ only the $j = k$ term survives, so

$$k! \, a_k = f^{(k)}(x_0).$$

25.4 The Weierstrass approximation theorem

Every continuous function on $[a, b]$ is almost a polynomial, in the following sense.

Theorem 25.6 [Weierstrass approximation theorem]. *If $f \in C([a, b])$, there is a sequence of polynomials $\{P_n\}$ with $P_n \to f$ uniformly on $[a, b]$.*

Reductions. It suffices to take $[a, b] = [0, 1]$: if polynomials $P_n \to f(a + (b - a)s)$ uniformly for $s \in [0, 1]$, then the polynomials $P_n\big(\frac{x-a}{b-a}\big) \to f(x)$ uniformly on $[a, b]$. It also suffices to take $f(0) = f(1) = 0$: for general $\tilde{f} \in C([0, 1])$, apply that case to $f(x) = \tilde{f}(x) - \tilde{f}(0) - x\big(\tilde{f}(1) - \tilde{f}(0)\big)$; if $P_n \to f$ uniformly, then $P_n(x) + \tilde{f}(0) + x\big(\tilde{f}(1) - \tilde{f}(0)\big) \to \tilde{f}(x)$ uniformly.

Idea. Build polynomials Q_n that behave like a "Dirac delta": nonnegative, total integral 1, and concentrating at 0. Then $P_n(x) = \int_0^1 f(t) Q_n(t - x) \, dt$ averages f over smaller and smaller neighborhoods of x.

Lemma 25.7 [Approximate identity]. *Let* $c_n = \left(\int_{-1}^{1}(1-x^2)^n\,dx\right)^{-1} > 0$ *and* $Q_n(x) = c_n(1-x^2)^n$. *Then*

1. $\int_{-1}^{1} Q_n = 1$ *for all* n;
2. $Q_n \geq 0$ *on* $[-1,1]$ *for all* n;
3. *for every* $\delta \in (0,1)$, $Q_n \to 0$ *uniformly on* $\{\delta \leq |x| \leq 1\}$; *more precisely* $0 \leq Q_n(x) \leq \sqrt{n}(1-\delta^2)^n$ *there, and* $\sqrt{n}(1-\delta^2)^n \to 0$.

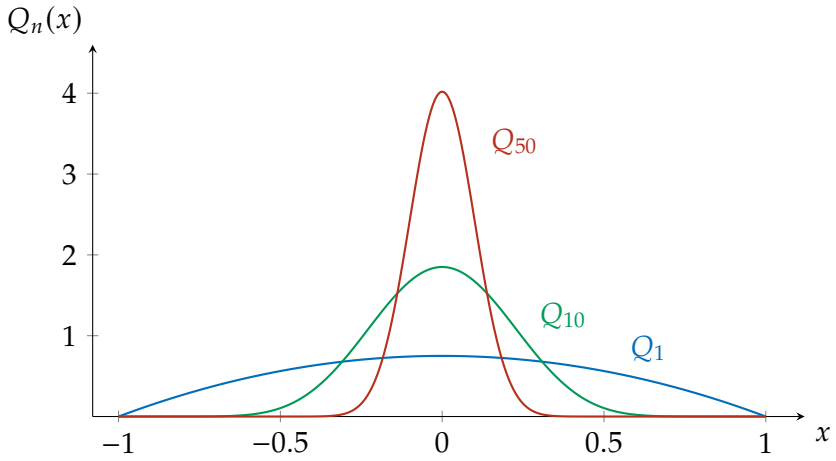


Figure 14: $Q_n(x) = c_n(1-x^2)^n$ for $n = 1, 10, 50$, with c_n computed exactly ($c_1 = 0.75$, $c_{10} \approx 1.85$, $c_{50} \approx 4.02$). Each has integral 1; as n grows the mass concentrates near 0.

Proof. (1) holds by the choice of c_n, and (2) is clear. For (3), first $(1-x^2)^n \geq 1 - nx^2$ on $[-1,1]$: the function $h(x) = (1-x^2)^n - (1-nx^2)$ is even, $h(0) = 0$, and $h'(x) = 2nx\big(1-(1-x^2)^{n-1}\big) \geq 0$ on $[0,1]$, so $h \geq 0$ by the mean value theorem. Hence

$$\begin{aligned}\frac{1}{c_n} &= 2\int_0^1 (1-x^2)^n\,dx \geq 2\int_0^{1/\sqrt{n}} (1-x^2)^n\,dx \geq 2\int_0^{1/\sqrt{n}} (1-nx^2)\,dx \\ &= 2\Big(\frac{1}{\sqrt{n}} - \frac{n}{3}n^{-3/2}\Big) = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}},\end{aligned}$$

so $c_n < \sqrt{n}$. For $\delta \leq |x| \leq 1$, $1 - x^2 \leq 1 - \delta^2$, so $0 \leq Q_n(x) \leq \sqrt{n}(1-\delta^2)^n$. Finally

$$\big(\sqrt{n}(1-\delta^2)^n\big)^{1/n} = (n^{1/n})^{1/2}(1-\delta^2) \to 1 - \delta^2 < 1,$$

so by the root test $\sum_n \sqrt{n}(1-\delta^2)^n$ converges and its terms tend to 0. The bound is independent of x, so the convergence is uniform. □

Proof of Theorem 25.6. Let $f \in C([0,1])$ with $f(0) = f(1) = 0$, and extend f by $f(x) = 0$ for

$x \notin [0,1]$; the extension is continuous on $\mathbb{R}$. Define

$$P_n(x) = \int_0^1 f(t)\, Q_n(t-x)\, dt = \int_0^1 f(t)\, c_n \big(1-(t-x)^2\big)^n\, dt.$$

Expanding, $c_n(1-(t-x)^2)^n = \sum_{k=0}^{2n} p_k(t) x^k$ for polynomials p_k, so $P_n(x) = \sum_k \big(\int_0^1 f(t) p_k(t)\, dt\big) x^k$ is a polynomial.

For $x \in [0,1]$, substituting $t = x + u$ (change of variables with $\varphi(u) = x + u$) and using $f(x+u) = 0$ for $u \notin [-x, 1-x]$ and $[-x, 1-x] \subset [-1,1]$,

$$P_n(x) = \int_{-x}^{1-x} f(x+u) Q_n(u)\, du = \int_{-1}^{1} f(x+u) Q_n(u)\, du.$$

By Lemma 25.7(1), $f(x) = \int_{-1}^1 f(x) Q_n(u)\, du$, so by $Q_n \geq 0$,

$$|P_n(x) - f(x)| \leq \int_{-1}^1 |f(x+u) - f(x)|\, Q_n(u)\, du.$$

Let $\epsilon > 0$. If $f \equiv 0$ then $P_n = 0 = f$; otherwise let $C = \max_{[0,1]} |f| > 0$, which exists by the min/max theorem. The extended f is continuous on the closed bounded interval $[-1,2]$, which contains every $x+u$ above, so it is uniformly continuous there: choose $\delta \in (0,1)$ with $|f(y) - f(z)| < \epsilon/2$ for $y, z \in [-1,2]$, $|y - z| \leq \delta$. By Lemma 25.7(3) choose M with $\sqrt{n}(1-\delta^2)^n < \frac{\epsilon}{8C}$ for $n \geq M$. Split the integral at $|u| = \delta$. For $n \geq M$ and every $x \in [0,1]$,

$$|P_n(x) - f(x)| \leq \frac{\epsilon}{2} \int_{|u| \leq \delta} Q_n(u)\, du + \int_{\delta \leq |u| \leq 1} 2C\sqrt{n}(1-\delta^2)^n\, du < \frac{\epsilon}{2} + 2C \cdot \frac{\epsilon}{8C} \cdot 2 = \epsilon,$$

using $\int_{|u|\leq\delta} Q_n \leq 1$ and that $\{\delta \leq |u| \leq 1\}$ has total length $2(1-\delta) < 2$. So $P_n \to f$ uniformly on $[0,1]$. □

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